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A Systematic Review of Endovascular Stent Grafting Versus Open Surgical Repair for the Treatment of Arch/Descending Thoracic Aortic Aneurysms

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A Systematic Review of Endovascular Stent Grafting Versus Open Surgical Repair for the Treatment of Arch/Descending Thoracic Aortic Aneurysms

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ABSTRACT

Aims

A thoracic aortic aneurysm (TAA) is an abnormal dilatation of the aorta at any part of its course from the heart to the diaphragm. If untreated, aneurysms will continue to expand with consequent increased risk of serious complications like dissection, rupture, or death. Our objective was to review comparisons of the effectiveness of endovascular stent grafting (ESG) against open surgical repair (OSR) for treatment of chronic arch or descending thoracic aortic aneurysms (TAA).

Methods and Results

Studies comparing ESG and OSR were sought; including randomised, quasi-randomised and non-randomised controlled trials, cohort studies and case-control studies matched on key outcomes. Patients had to have received elective treatment for arch/descending thoracic aortic aneurysms. MEDLINE, EMBASE, CENTRAL, WHO International Clinical Trials Routine data collection, current controlled trials, clinical trials, and the NIHR portfolio were searched.

Five comparative cohort studies met inclusion criteria, reporting a total of 3955 ESG and 21,197 OSR patients. Meta-analysis of unadjusted short-term all-cause mortality favoured ESG (odds ratio: 0.75; 95 %CI (0.55-0.1.03)). Heterogeneity was identified between the larger and smaller studies. Meta-analysis of adjusted short-term all-cause mortality favoured ESG odds ratio 0.71, 95%CI (0.51, 0.98), no heterogeneity present. Long-term survival from all-cause mortality favoured OSR in larger studies and ESG in smaller studies. Freedom from re-interventions in the long-term favoured OSR. Studies reporting short-term non-fatal complications suggest fewer events following ESG although limited data prevented meta-analysis.

Conclusion

There is limited and increasingly dated evidence on the comparison of ESG and OSR for treatment of arch/descending thoracic aortic aneurysms.

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KEY WORDS

Systematic Review, Meta-analysis, Thoracic Aortic Aneurysm, Open Surgical Repair, Endovascular Stent Graft

STRENGTHS AND LIMITATIONS OF THIS STUDY

Strength:

- To the authors knowledge this is the first systematic review to investigate two-armed studies looking only at ESG versus OSR for treatment of chronic arch or descending thoracic aortic aneurysms. Other reviews include dissections, ruptures and treatment for other conditions.
- This systematic review also considers evidence from both randomized and non-randomized study designs.

Limitation:

- Statistical analysis of the non-fatal complications was not possible due to the limited and inconsistent reporting of such complications.

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INTRODUCTION

Rationale

A thoracic aortic aneurysm (TAA) is an abnormal expansion of the aorta at any part of its course from the heart to the diaphragm. Aneurysms manifest with acute symptoms or they may be chronic. If untreated, TAA can continue to expand and the annual risk of aneurysm rupture increases from 2% in aneurysms smaller than 5cm in diameter to 7% in aneurysms over 6cm in diameter[1]. The majority of aneurysms (and therefore studies) centre on the ascending or abdominal segments of the aorta, arch and descending aneurysms are relatively under-studied. Therefore this review focuses on chronic aneurysms affecting the arch or descending thoracic aorta.

There are currently two main methods of repair for arch/descending TAA: Open Surgical Repair (OSR) and Endovascular Stent Grafting (ESG). OSR involves replacing the aneurysmal segment of aorta with a surgical graft during open surgery using heart-lung bypass. As the arch and descending aorta supply blood to the brain and spinal cord, OSR is high risk, with reported mortality and paraplegia rates of approximately 5% and 10% [2] respectively. In ESG a stent is placed across the aneurysm, providing a new conduit through which the blood flows. ESG is most often selected for patients who are deemed at high risk of death or permanent disability from OSR. However, some patients may not have a suitable anatomy/morphology for ESG procedures [2]. Concerns have also been raised about long-term durability and safety of the implanted devices [3 4] due to complications such as stent migration or endoleaks, with the incidence of late endoleaks (leaks around or through the stent graft that occur more than year after insertion) reported as being between 2.5% and 27.8% [5]. The need for re-intervention due to technical failures of the stent following ESG is also high (23% at 5 years) [6 7]. With each re-intervention there is an added risk of complication either due to the increased complexity of the procedure or deteriorating health of the patient [2].

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One previous systematic review comparing only elective OSR and ESG for descending and arch TAAs has been reported [8]. This systematic review [8] was restricted to data from randomized control trials in which patients were randomly assigned to ESG and OSR, but no eligible trials were identified when searching relevant data based up to Jan 2016. There are other systematic reviews [9 10] that have included non-randomized studies; however, these reviews also included emergency cases and other thoracic aortic pathologies (e.g. dissection or rupture). As such, they may not represent a true comparison of elective ESG and OSR procedures for descending thoracic aortic aneurysms. Given the lack of RCT evidence, this systematic review includes non-randomized comparative studies of the relative effectiveness of OSR or ESG.

The decision to operate requires a favourable balance between the risks and benefits of that intervention compared to those attached to the expected outcomes of the disease. This balance is in turn influenced by a number of factors. Aneurysm size correlates to the risk of death (if left untreated) and so patients are usually referred once the aneurysm crosses a chosen threshold. At that point, patient fitness (*and life expectancy*), the reality or possibility of an intrinsic weakness affecting the whole aorta, the technical feasibility of the proposed intervention and patient wishes, come into play. This systematic review takes a step back from these issues and examines only the outcomes for patients in whom a decision to intervene has been made. This review is an essential starting point for the analysis of effective treatments for thoracic aortic aneurysms.

Objectives

The aim of this systematic review was to assess the effectiveness of endovascular stent grafting (ESG) compared to open surgical repair (OSR) as a treatment for chronic arch or descending TAA.

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METHODS

Protocol and registration

The review protocol was registered in the PROSPERO database (CRD42017054565)[11]

Eligibility criteria

Study designs

All identified studies that directly compared ESG and OSR, including RCTs, quasi-randomised and non-randomised controlled trials, comparative cohort studies and case-control studies matched on main outcomes were sought [12].

Non-randomised studies that controlled for selection bias were included – irrespective of study size. Acceptable 'anti-selection bias" methods included matching patients at baseline for important covariates, matching patients using propensity scores or adjustment for important covariates during analysis. Non-randomised studies that did not use these methods but included larger sample sizes (more than 250 patients) were also included as they have the potential to reduce sampling variation relative to smaller unmatched studies[13].

Non-English language papers were noted and, where possible, data were extracted from the abstract. No translation of full text articles was possible.

Study Participants

Study participants had to receive elective treatments for arch and, or descending thoracic aortic aneurysms. Where emergency interventions or other thoracic aortic conditions (such as intramural haematoma, rupture, and dissection) were reported, studies were excluded, unless results for patients receiving elective treatment for arch/descending TAA were reported separately. Studies that reported secondary (re-intervention) procedures for arch/descending TAAs were included only if the original, primary procedure for each participant could be identified.

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Study interventions

Open Surgical Repair was defined as replacement of the aneurysmal aorta with a prosthetic conduit via a sternotomy or thoracotomy with circulatory support. Endovascular stent grafting was defined as transluminal introduction of a stent-graft under X-ray guidance.

Outcome measures

This review assessed short-term and long-term clinical effectiveness. Short-term was defined as either within 30-days of the intervention and/or time until discharge following the intervention. Since most studies did not report long-term outcomes conditional on surviving the short-term period, long-term analyses covered the time from intervention until the end of follow-up. Primary outcomes were all-cause mortality and aneurysm-related mortality. Secondary outcomes were re-interventions and non-fatal complications. The latter were sub-classified as vascular, neurological, cardiac, respiratory and 'other' complications.

Information sources

MEDLINE, EMBASE, CENTRAL, WHO International Clinical Trials Routine data collection, Current Controlled Trials, Clinical Trials, and the UK National Institute for Health Research portfolio were searched. The search strategy (Appendix 1) was adapted as appropriate for each database (for example to allow for variations in controlled vocabulary terms and syntax [14]). Electronic searches identified publications between 1994, date of first reported case of ESG for descending and arch TAA[15], and March 2020.

Websites of regulatory bodies and Health Technology Assessment agencies were screened for unpublished relevant reports. A citation search of all included studies was conducted via Web of Science. Studies from systematic reviews that satisfied the inclusion criteria of this review were screened for eligibility.

Study selection

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All titles and abstracts were screened by one reviewer (AM). A second investigator (JG) reviewed 10% of all titles and abstracts classed as eligible or not eligible and all titles and abstracts classified as unclear by the first reviewer. The full text of selected studies was assessed by one investigator (AM) and classified as relevant, not relevant or unclear according to the inclusion criteria. A second investigator (JG) reviewed 10% of all papers, as well as all studies deemed unclear. Disagreements at either stage were resolved by discussion and where necessary a third reviewer (PM) arbitrated. Where required, a clinical expert (PS) was asked to screen papers.

Data collection process

A data extraction form was developed based on The Cochrane Collaboration data collection form [16]. Data were extracted by one reviewer (AM) and checked by a second reviewer (JG) for accuracy. Disagreements were resolved by discussion or arbitration by a third reviewer (PM).

Data items

Reported short-term and longer-term all-cause mortality, as well as short-term and longer-term aneurysm-related mortality were collected. All reported complications were recorded and grouped according to expert clinical guidance as described above as neurological, cardiac, respiratory or 'other' complications, with endoleaks following ESG recorded separately.

Risk of bias in individual studies

The risk of bias was assessed using the ROBINS-I tool [17] designed for non-randomised studies. The ROBINS-I tool classifies the risk of bias in seven domains (*confounding; selection of participants; classification of interventions; deviations from intended interventions; missing data; measurement of outcomes; selection of the reported results*) and overall as either critical, severe, moderate, low, or unclear.

Synthesis of results

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Point estimates and precision of the incidence of short and long-term events and unadjusted and adjusted comparisons (odds ratios (OR) hazard ratios (HR)) were recorded. Meta-analysis of short-term event-comparisons was based on the OR with 95% confidence interval (CI) and conducted using RevMan [18]. An overall treatment effect was calculated for the log (OR) of each outcome using a (normal) random effects model. Where comparisons could not be made, data were tabulated and reported narratively.

Heterogeneity between studies was assessed according to Cochrane guidelines, on clinical criteria, by visual inspection of plots of the data, from the chi-square test for heterogeneity, and the I-squared statistic. An I^2 statistic of 25-50% indicated low heterogeneity; 50-75% indicated moderate heterogeneity, and over 75% indicating significant heterogeneity.

RESULTS

Study selection

The search identified 5163 unique studies. Of these, 5078 studies were not eligible. Ineligibility was primarily because of irrelevant anatomy (ascending, abdominal, or peripheral aneurysms), irrelevant pathology (acute dissection or other acute aortic syndromes), or because the study pooled data from a mixture of anatomical and pathological situations such that data on chronic arch and DTA aneurysms could not be extracted with confidence. Thus 85 papers underwent full text screening. 75 papers were rejected at this stage leaving 10 papers to be included in the full review. Six papers reported data from the Gore TAG Trial [19-24], leaving five unique comparative studies that fulfilled inclusion criteria. Further details of the selection process are shown in the PRISMA diagram (Figure 1).

Brief summaries of the identified studies are given in Table 1, with detail provided in Appendix 2. Overall, the five studies included data on 3955 ESG and 21,197 OSR patients and all studies were conducted in Europe or the USA. However, two studies had non-overlapping or only partially

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overlapping recruitment periods between the ESG and OSR intervention groups within their respective study (Gore Tag and Piffaretti et al 2007[19 25]). There were clear differences in patient characteristics between the two treatment groups. ESG patients were consistently older, with average differences ranging from 2 to 9 years in all but one analysis (Goodney et al 2011 the propensity matched cohort). Similarly, in most studies the proportion of men was greater in the ESG arm.

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Table 1: Study characteristics

Study	**Von Allmen et al 2014**	**Andrassy et al 2011**	**Goodney et al 2011**	**Goodney et al 2011 (propensity matched cohort)**	**Gore TAG**	**Hughes et al 2014**	**Piffaretti et al 2007**
METHODS							
Study Design	Comparative cohort	Comparative cohort	Comparative cohort	Comparative cohort	Comparative cohort	Comparative cohort	Comparative cohort
Method of matching	None	None	None	Stratified propensity score	ESG patients judged eligible for OSR	None	OSR patients judged eligible for ESG
PARTICIPANTS							
Country	UK	Germany	USA	USA	USA	USA	Italy
Setting	Multi-centre – Hospital episode statistics	Single centre	Multi-centre – Medicare data	Multi-centre – Medicare data	Multi-centre	Multi-centre	Single-centre
Time frame	2006 to 2011	ESG 1997-2008,	1998-2007	2003-2007	ESG 1999- 2001,	1998-2007	ESG 2000-2007,

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		OSR 1992-2002			OSR pre 1999 (n=44) & 1999-2001 (n=50)		OSR 1996-2000
Recruitment	Total 618 ESG 354 OSR 264 OSR	Total 77 ESG 53 OSR 24	Total 13,998 ESG 2433 OSR 11,565	Total 1100 ESG 550 OSR 550	Total 234 ESG 140 OSR 94	Total 8967 ESG 712 OSR 8255	Total 28 ESG 17 OSR 11
Age summary	median (IQR) ESG 73(66,78), OSR 71 (63-76)	mean (SD) ESG 70(10), OSR 67(12)	mean (95% C.I.) ESG 75.9 (75.6, 76.1), OSR 73.8 (73.7, 73.9)	mean (95% C.I.) ESG 71.1 (70.8, 71.44), OSR 70.7 (70.7, 71.1)	mean(SD) ESG 75.9 (10.4), OSR 68.2 (10.2)	median (IQR) ESG 72 (63- 78), OSR 63 (52, 72)	mean (SD) ESG 66 (10), OSR 61 (13)
Proportion male	ESG 232 (65.5%), OSR 137 (51.9%)	ESG 32 (60%), OSR 13 (54%)	% (95% C.I.): ESG 58.7 (56.7, 60.7), OSR 55.4 (54.8, 56.3)	% (95% C.I.): ESG 64.0 (59.9, 68.0), OSR 68.1% (64.2, 72.0)	ESG 90(57%), OSR 48 (51%)	5415 (61.3%) ESG, 437 (65.6%) OSR	ESG 14 (82%), OSR 8 (73%)
OUTCOMES							
Short-term	Death from any	Death from any	Death from any	Death from any	Death from any	Death from any	Death from any

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	cause Re-intervention All-cause mortality (unadjusted	cause Re-intervention Complications	cause	cause	cause Complications	cause Complications	cause Complications
Long-term	Time to death from any cause Time to aortic-related death Time to aortic-related re-intervention	Time to death from any cause	Time to death from any cause	Time to death from any cause	Time to death from any cause Time to aneurysm-related death aneurysm-related mortality Time to re-intervention Time to major adverse event		Time to death from any cause

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Risk of bias within studies

One study (Goodney et al 2011) [26] was judged to be of overall moderate risk of bias and four studies were at severe risk of bias due to the potential for confounding (Appendix 3).

Two large cohorts derived from UK and USA administrative databases (Von Allmen et al 2014 and Hughes et al 2014) [27 28] did not attempt to address selection bias at the design stage, but did adjust for some confounders in the analysis. A third study (Goodney et al 2011), using USA administrative data, included both risk-factor-adjusted models and analysis of a subset of the data defined by propensity score matching. Propensity score methods rely on all of the important confounders being available and included[29]. As this study did not include all confounders considered important, such as aneurysm size, this study was judged to have moderate risk of bias due to confounding.

One study (Gore TAG)[19] included only ESG patients who would also have been eligible for OSR and the another study (Piffaretti et al 2007) [25] included only OSR patients who would be suitable for ESG, both according to clinician judgement. Since suitability was decided retrospectively, such matching may not be robust.

A further potential bias with each study was selective reporting. Although the measurements and analyses of the outcomes were clearly defined within each study, no evidence of a pre-study analysis plan was identified.

Synthesis of results

Short-term death from any cause (30-day or in hospital)

Unadjusted Mortality:

All five studies [19 25-28] reported unadjusted 30 day or in-hospital mortality (figure 2). Overall, the odds of short-term mortality were lower in the ESG group than the OSR group (OR: 0.75; 95 %CI. (0.55-1.03)). Moderate heterogeneity was present in the overall analysis (I^2= 40%), mainly due to

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differences between the two smaller studies [19 25] and the three larger studies [26-28], with a large I^2 for subgroup differences between the larger studies subgroup and the smaller studies subgroup of 84.5%.

For the three larger studies[26-28] (3798 ESG and 21,092 OSR patients), the treatment effect for death was smaller (OR: 0.84 95%CI (0.72, 0.98)) than that for the two smaller studies [19 25](157 ESG and 105 OSR patients, combined OR: 0.18 95%CI (0.06, 0.59)). There was no evidence of heterogeneity in results between these subgroups (I^2=0).

Adjusted Mortality:

The three larger studies [26-28] included adjusted analysis of short-term mortality (Figure 3). One study used stratified propensity matching. Another study [27] adjusted for preoperative comorbidities, independent predictors of mortality and post-operative complications and the third study adjusted for age and sex. The OR for ESG, relative to OSR, for these three studies combined was 0.71, 95%CI (0.51, 0.98). No evidence of heterogeneity was identified in this meta-analysis (I^2=0%)

All-cause mortality (following hospital discharge or after 30 days)

Meta-analysis of the longer-term all-cause mortality was not possible as follow-up varied between the four studies that reported this outcome (Gore Tag, Goodney et al 2011, Piffaretti et al 2007 and Von Allmen 2014) [19 25 26 28] . The results in the larger studies (Goodney et al 2011, Von Allmen 2014)[26 28] differed from those in the smaller studies (Gore Tag and Piffaretti et al 2007)[25 30]. We note that the analyses included the short-term deaths as well as those occurring beyond hospital discharge or after 30 days.

Von Allmen et al 2014 reported time to death from any cause up to five years, throughout which OSR had higher overall survival. Adjusting for age and sex, ESG had higher odds of death at one year

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(OR: 1.10 95%CI (0.70, 1.73); p=0.667). Additionally, the hazard ratio for ESG relative to OSR up to five years, adjusting for age and sex, was HR=1.45 (95%CI [1.08, 1.94], p=0.013).

Goodney et al 2011 also reported time to death from any cause up to five years. Adjusting for age, sex, race, era of procedure and Charlson comorbidity score, five-year survival for OSR was higher than for ESG (89% versus 79%, log-rank p<0.0001). In the propensity-matched cohorts, five-year life-table analysis reported higher survival in the OSR group (81% 95%CI (77%-85%)) than for the ESG group (73% 95%CI (68%, 76%)) (log-rank p=0.007).

Time to death from any cause in the Gore TAG study did not differ over five years between the ESG and OSR groups with survival probability at five years in both groups was 63% (log rank P value= 0.625). Piffaretti et al 2007 reported survival overall in the ESG group up to five years and the OSR group up to 13 years. For the five years in which survival is reported in both groups, ESG had a higher survival compared to OSR although this was based on only 4 deaths, and the follow-up periods differed substantially (log rank p>0.05).

Aneurysm/Aortic-related mortality

Meta-analysis of Aneurysm/aortic-related mortality could not be conducted. No studies reported aneurysm-related mortality during hospitalization or up to 30 days separately to the long-term. Von Allmen et al 2014 reported aortic-related mortality up to five years. Time to aortic-related death was did not differ between OSR group than the ESG group during the five-year follow-up (log-rank p=0.270). This study[28] also reported adjusted (for age and sex) five year HR of 1.27 (0.81 to 1.98; P =0.298) for ESG patients relative to OSR patients. Gore TAG[19] reported aneurysm-related mortality up to five years, with longer time to aneurysm-related death in the ESG group over this period (log-rank p= 0.012).

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Short-term re-interventions

One study reported this outcome (Von Allmen et al 2014[28]). This study reported aortic-related re-interventions in 21 (5.9%) ESG patients and 4 (1.5%) OSR patients, an adjusted OR of 2.97 (95%CI [1.09, 8.10], p=0.033) for ESG versus OSR.

Long-term re-interventions

Two studies reported this outcome (Von Allmen et al 2014[28] and Gore TAG [19]). Von Allmen et al 2014[28] reported time to aortic-related re-interventions up to five years. Time to re-intervention was longer in the OSR group compared to the ESG group (log-rank p=0.007), with adjusted (for age and sex) hazard ratio at five years of 1·70 (95%CI [1·11 to 2·60], p =0·014). In Gore TAG[19], time to re-intervention over five years was longer in the ESG group (log-rank P=0.011). We note that these analyses included the short-term re-interventions.

Complications

Table 2 summarises the short-term complications reported in all five studies[19 25-28]. Most studies reported a small number of neurological, pulmonary, cardiac and other complications. However, one larger study (Hughes et al 2014)[27] reported lower odds for ESG patients for neurological complications, OR of 0.48 (95%CI (0.26, 0.86), p=0.015), pulmonary complications, OR: 0.38 (95%CI (0.21, 0.67), p=0.001) and cardiac complications, OR: 0.24 (95%CI (0.15, 0.37), p<0.001). The Gore TAG study[19] reported endoleak rates for ESG patients of 8.5%.

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Table 2: Short-term complications

Study	Gore TAG		Piffaretti et al 2007		Hughes et al 2011	
Arm (participants)	**ESG (n=140)**	**OSR (n=94)**	**ESG (n=17)**	**OSR (n=11)**	**ESG (n=712)**	**OSR (n=8255)**
NEUROLOGICAL						
Paraplegia	NR	NR	NR	NR	NR	NR
Paraplegia/ Paraperisis	4 (3%)	13 (14%)	NR	NR	NR	NR
Cerebral Vascular Accident	5 (4%)	5 (4%)	2 (9%)	1 (12%)	NR	NR
Neurological-unspecified type	NR	NR	NR	NR	20 (2.8%)	273 (3.3%)
RESPIRATORY						
Respiratory failure	5 (4%)	19 (20%)	NR	NR	NR	NR
Pneumonia	NR	NR	2 (12%)	3 (27%)	NR	NR
Pulmonary-unspecified type	NR	NR	NR	NR	17 (2.4%)	462 (5.6%)
CARDIAC						
Myocardial Infarction	0 (0%)	1(1%)	0 (0%)	1 (9%)	NR	NR
Cardiac-unspecified type	NR	NR	NR	NR	28 (2.9%)	1252 (15.2%)
ENDOLEAKS						
Endoleaks	12 (8.5%)	NR	NR	NR	NR	NR
OTHER						
Peripheral vascular disease	20 (14%)	4 (4%)	NR	NR	NR	NR

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Renal failure	2 (1%)	12 (13%)	NR	NR	NR	NR
Wound infection/dehiscence	5 (4%)	10 (11%)	NR	NR	NR	NR
GI complication	3 (2%)	6 (6%)	NR	NR	NR	NR
Post-implant syndrome	NR	NR	4(18%)	0(0%)	NR	NR
Note: Goodney et al 2011 and Von Allmen et al 2014 did not report short-term complications						

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DISCUSSION

This is the first systematic review that reports results from non-randomized comparative studies regarding the effectiveness of elective ESG and OSR for arch and or descending TAA. Previous reviews have either included single arm studies only[31] or were comparative reviews that included non-elective procedures[9 32], or mixed pathologies such as dissections[10 33]. The results of these reviews may not be applicable to patients undergoing elective procedures for descending or arch TAA pathology.

In this review only five comparative cohort studies were identified. The lack of RCTs comparing ESG and OSR for elective arch/descending thoracic aortic aneurysms may reflect the lack of equipoise from practitioners and patients. Well-designed non-randomized trials can demonstrate a high degree of concordance with the results of well-designed RCTs, provided that all confounders are adequately accommodated in the design and/or analysis[34]. However, the studies identified in this review had moderate to serious risk of bias, mainly due to potential for confounding.

Although, complications in the short-term also appear to favour ESG, evidence regarding short-term complications is either lacking or inconsistently reported. All five studies[19 25-28] suggested that the less invasive nature of ESG confers lower short-term mortality and fewer short-term non-fatal complications than OSR. The two small unadjusted studies[19 25] had a much larger effect than larger studies[26-28]. These larger studies used more robust methods for addressing confounding; this may

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indicate the presence of publication bias where only smaller studies with larger effect sizes may have been published.

Longer-term results are difficult to interpret since most results fail to separate early risk from longer term survival. The two smaller studies[19 25] reported continuing survival benefit for ESG patients in the long-term but are limited by small numbers, different recruitment periods, and failure to exclude early risk. The survival advantage associated with ESG in the Gore TAG study disappears by year five, which suggests that the hazards are lower for OSR patients that survive the initial procedure. Conversely, the two larger studies (Von Allmen et al 2014 and Goodney et al 2011[26 28]) report that the short-term survival advantage of ESG are reversed by twelve months, and that the OSR group has an increasing survival advantage thereafter. Similarly, time to re-intervention favoured ESG in the smaller Gore TAG study whilst time to aortic-related re-interventions favoured OSR in the larger study Von Allmen et al 2014[28].

The evidence regarding mortality is inconsistent across the studies with more recent, larger studies[26-28] reporting a more modest survival benefit for ESG in the short-term and a survival benefit for OSR in the long-term. The two[19 25], smaller studies suggest a large reduction in the risk of mortality and short term complications in the short run and a survival benefit in the long-run for ESG.

Inconsistency of results across studies may be due to heterogeneity of the study populations, both within and across studies. Differences in recruitment periods may reflect changes in technology and practice in both groups. The larger studies [26-28] were based on patients recruited up to twenty years later than the smaller studies. Outcomes may have improved differentially with OSR patients receiving proportionally better outcomes due to general improvements in surgical practice and patient selection over time[35]. A potential reason for the reported long-term survival benefit of OSR in more recent, larger studies may be differences in patient risk in each intervention group. Open

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surgical repair is advocated for younger, lower-risk patients, and EVAR is preferred for older, higher-risk patients[36]. This factor may not have been adequately accommodated in adjusted analyses.

The three large studies[26-28] were based on administrative health records with case ascertainment that depends on accurate coding, which can vary across and within providers[37]. Furthermore, one of the small studies, Piffaretti et al 2007[25], recruited patients from one centre which may have compromised external validity as they may for example, depend on the expertise within a centre and the nature of the patients that centre treats[30 38]. Additionally, the Gore TAG study[19] utilised a more restricted inclusion criteria for ESG patients compared to those who received OSR, potentially introducing heterogeneity between the two groups.

The headline conclusion of this review is that whilst there is a degree of variability, OSR is better for long-term survival and avoidance of re-intervention, and that ESG is better for early outcomes. This conclusion is based on non-randomised comparisons in which the influence of important confounders for which patients have ESG or OSR has not been adjusted for. Deeper exploration is required, but randomised data may be difficult to acquire as there are well-established guidelines about which patients are favoured for OSR and which for ESG. Thus, to compare ESG and OSR it is first necessary to establish an understanding of the reasons driving clinical management, for example by a structured consensus analysis such as a Delphi study. Thereafter, unless there is clinical equipoise about which patients are best served by OSR or ESG, the best quality evidence to establish the relative merits of ESG and OSR for arch/ DTA aneurysms is likely to be a systematic, prospective, observational, comparative study.

Limitations

The findings in this review are limited by the number and potential for confounding of the five studies[19 25-28] identified, so that definitive conclusions are not possible. The adjustment for confounders is likely to be only partial and some bias in the estimates of differences in mortality and complications may persist, despite attempts to reduce bias in large studies. Statistical analysis of the

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non-fatal complications was not possible due to the limited and inconsistent reporting of such complications.

CONCLUSIONS

This systematic review is the first to consider evidence from non-randomised studies that directly compare ESG and OSR for the treatment of elective arch/descending thoracic aortic aneurysms. Other reviews conducted in this area have been complicated by the inclusion of treatments for emergency cases or other thoracic aortic pathologies (e.g. thoracic aortic ulcer, or dissection).

There is limited and increasingly dated evidence addressing the comparison of ESG and OSR in the management of elective arch and or descending thoracic aortic aneurysms. Further, high quality, evidence is required. Although large RCTs may not be possible in this relatively rare condition, future studies should use strong methodological study design in order to control for potential confounders and reduce potential biases. Additionally, given the current conflicting evidence, it is important that comparisons of OSR and ESG are conducted in both the short-term and the long-term.

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CONFLICT OF INTEREST

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AUTHOR CONTRIBUTIONS

Author contributions according to the CrediT (contributor roles taxonomy). Conceptualization- AM, JG, PS, LS, LV, AC, PM, SL; Data curation- AM ;Formal Analysis- AM, JG, PM; Funding acquisition- JG, AC, LV, CF, PS, LS, SL; Investigation - AM, JG, PM, LV, PS; Methodology - AM, JG, PM, LV, AC, CF, PS, LS, SL; Supervision - JG, PM, LV; Validation – AM, JG, PM, PS, LS, PC, SL; Writing – original draft - AM, JG, PM; Writing – review & editing - AM, JG, PM, AC, LV, CF, PS, LS, PC, SL.

DATA SHARING AGREEMENT

Data was extracted from the studies included in this Systematic Review and referenced below.

PUBLIC PATIENT INVOLVEMENT

No patient involved

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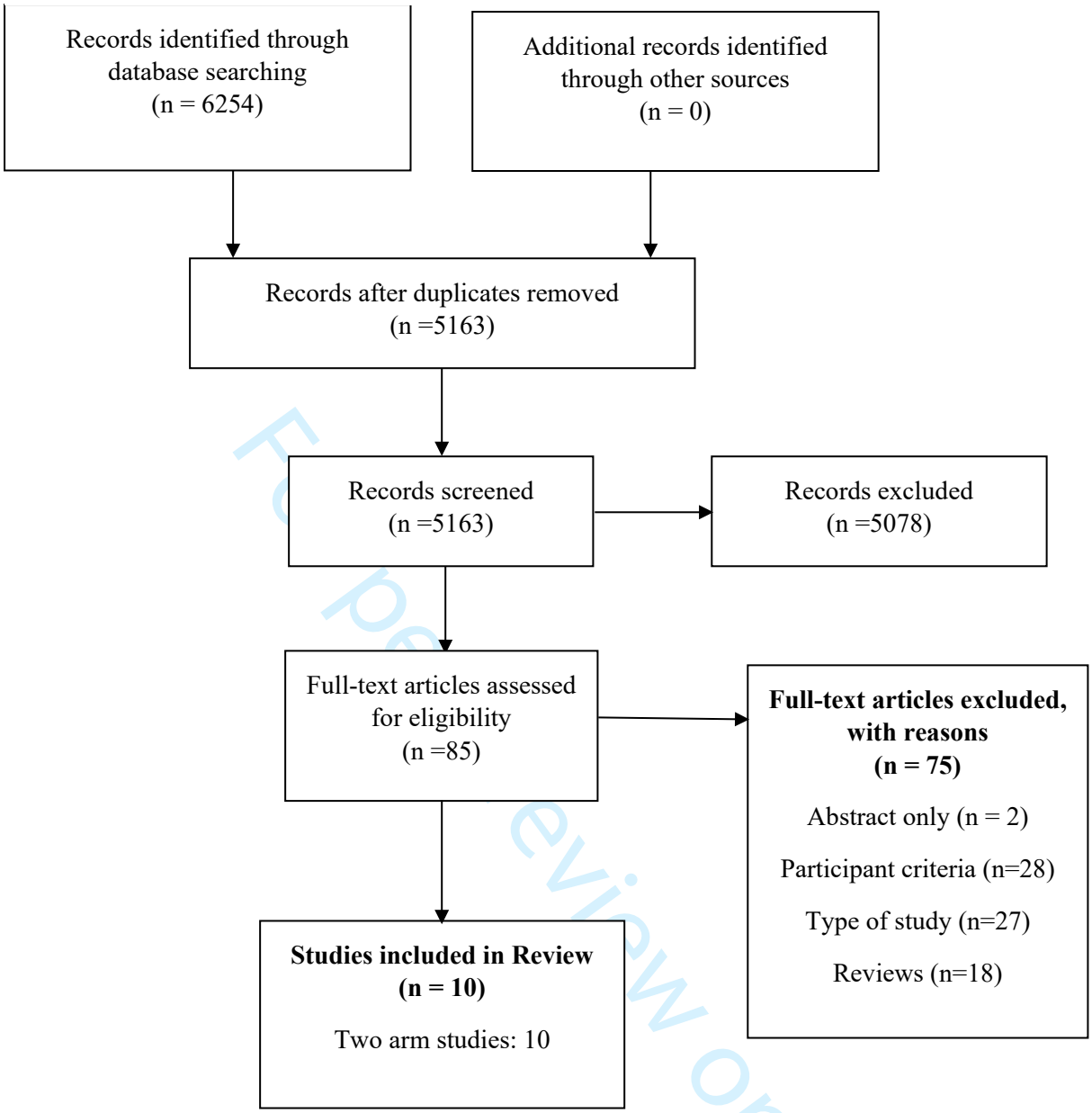
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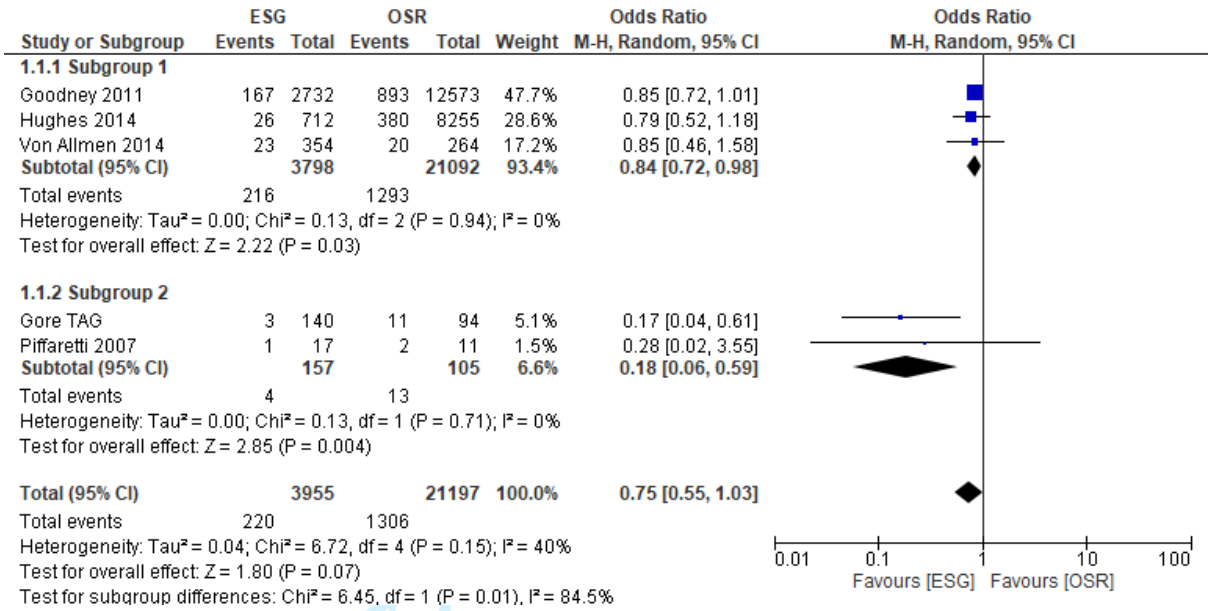
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Figure 1: PRISMA Diagram



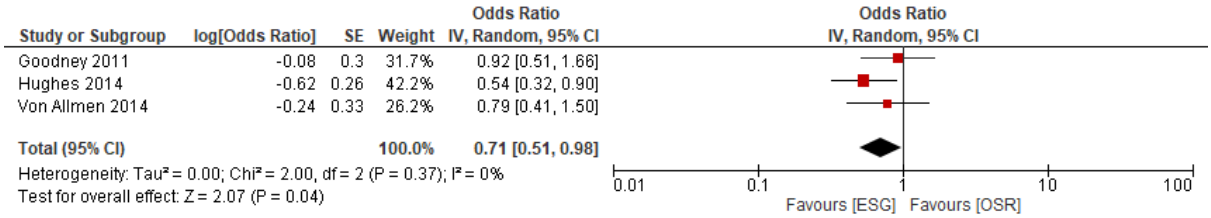
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Figure 2: Forest plot of unadjusted short-term all-cause mortality



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Figure 3: Forest plot of adjusted short-term all-cause mortality



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Appendix 1: Search Strategy

Medline/Central/HTA

1. (TEVAR or EVAR).ti,ab.
2. (endovascular adj1 (stent* or repair* or treat* or surg* or (aneurysm? adj1 repair*))).ti,ab.
3. (open adj2 (surg* or repair*)).ti,ab.
4. endoprosthe*.mp.
5. *Aortic Aneurysm, Thoracic/su [Surgery]
6. *Aortic Thoracic/su [Surgery]
7. Thoracic Surgical Procedures/
8. or/1-7
9. exp Aortic Aneurysm, Thoracic/
10. (thorac* adj2 aneurysm*).ti,ab.
11. (descending adj1 aort* adj1 aneurysm*).ti,ab
12. TAA*.ti,ab.
13. (arch adj3 aneurysm*).ti.ab.
14. or/9-13
15. morbid*.mp.
16. mortalit*.mp.
17. outcome?.mp.
18. complication*.mp.
19. (rupture? or endoleak?).mp.
20. device migration.mp.
21. (ae or co or de).fs

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22. (safe or safety or side effect* or undesirable effect* or treatment emergent or tolerability or toxicity or adrs or (adverse adj2 (effect or effects or reaction or reactions or event or events to outcome or outcomes))).ti,ab.
23. ((technical or surg*) adj2 (advance* or improve*)).mp.
24. (quality adj2 life).mp.
25. Mortality/
26. Mortbidity/
27. Postoperative complications/
28. Intraoperative complications/
29. Aortic Aneurysm, Thoracic/co
30. Conversion to Open Surgery/
31. “Quality of Life”
32. or/15-31
33. 8 and 14 and 32
34. (“clinical trial” or “clinical trial, phase I” or “clinical trial, phase ii” or clinical trial phase iii or clinical trial, phase iv or controlled clinical trial or “multicentre study” or “randomized controlled trial”).pt. or double-blind method/ or clinical trials, phase iii as topic/ or clinical trials, phase iv as topic/ or controlled clinical trials as topic/ or randomized controlled trials as topic/ or early termination of clincal trials as topic/ or multicenter studies as topic/ or ((randomi?ed adj7 trial*) or (controlled adj3 trial*) or (clinical adj2 trial*) or ((single or doubl* or tripl* or treb*) and (blind* or mask*))).ti,ab,kw. or (“4 arm” or “four arm”).ti,ab,kw.
35. cohort studies/ or longitudinal studies/ or follow-up studies/ or prospective studies/ or retrospective studies/ or cohort.ti,ab. or longitudinal.ti,ab. or prospective.ti,ab. or retrospective.ti,ab. or quasi-experiment.ti,ab.

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36. Clinical Trial.pt. or random*.tw. or Clinical Trials/ or clinical trial.tw. or Follow-up Studies/ or prospectiv*.tw. or control*.tw.
37. or/34-36
38. 33 and 37

EMBASE

1. (TEVAR or EVAR).ti,ab.
2. (endovascular adj1 (stent* or repair* or treat* or surg* or (aneurysm? adj1 repair*))).ti,ab.
3. (open adj2 (surg* or repair*)).ti,ab.
4. endoprosthe*.mp.
5. Thoracic Aorta Aneruysm/su [Surgery]
6. Thoracoabdominal aorta aneurysm/su [Surgery]
7. Thoracic Aorta/su [Surgery]
8. exp Thoracic aorta surgery/
9. or/1-8
10. exp Thoracic Aorta Aneurysm/
11. exp Thoracoabdominal aorta aneurysm/
12. Thoracic Aorta/
13. (thorac* adj2 aneurysm*).ti,ab.
14. (descending adj1 aort* adj1 aneurysm*).ti,ab.
15. TAA*.ti,ab.
16. (arch adj3 aneurysm*).ti,ab.
17. or/10-16
18. (outcome? or mortalit*).mp.
19. complication*.mp.
20. ((technical or surg*) adj2 (advance* or improve*)).mp.

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21. (quality adj2 life).mp.
22. Mortality/
23. Postoperative complication/
24. peroperative complication/
25. Thoracic Aorta Aneurysm/co
26. “Quality of Life”/
27. or/18-26
28. (Randomised Controlled Trial or Pragmatic Clinical Trial).pt or exp Randomized Controlled Trials as Topic/ or “Randomized Controlled Trial (topic)”/ or Randomized Controlled Trial/ or Randomization/ or Random Allocation/ or Double-Blind Method/ or Double Blind Procedure/ or Double-Blind Studies/ or Single-Blind Method/ or Single Blind Procedure/ or Single-Blind Studies/ or Placebos/ or Placebo/ or (random* or sham or placebo*).ti,ab,hw,kf,kw. or ((singl* or doubl*) adj (blind* or dumm* or mask*)).ti,ab,hw,,kw. or ((tripl* or trebl*) adj (blind* or dumm* or mask*)).ti,ab,hw,,kw.
29. random:.tw. or placebo:.mp. or double-blind:.mp.
30. controlled clinical trial/ or clinical trial/ or exp controlled study/ or randomized controlled trial/ or exp quasi experimental study/
31. or/28-30
32. 9 and 17 and 27 and 31
33. limit 32 to (human and yr=”1994-Current”)
34. exp cohort analysis/ or exp longitudinal study/ or exp prospective study/ or exp follow up/ or cohort*.tw.
35. limit 34 to (human and yr=”1994-Current”)
36. 9 and 17 and 27 and 35
37. 36 not 33
38. 33 or 37

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Appendix 2: Data extraction table

Study	**Von Allmen et al 2014**	**Goodney et al 2011**	**Goodney et al 2011 (propensity matched cohort)**	**Gore TAG**	**Hughes et al 2014**	**Piffaretti et al 2007**
METHODS						
Study Design	Comparative cohort	Comparative cohort	Comparative cohort	Comparative cohort	comparative cohort	Comparative cohort
Method of matching	None	None	Stratified propensity score	ESG patients judged eligible for OSR	None	OSR patients judged eligible for ESG
Inclusion Criteria	Hospital diagnosis code 171.2 (TAA without mention of			Fusiform descending thoracic aortic aneurysm at least twice the size of the	Nationwide inpatient Sample database of	Elective, Chronic fusiform or saccular aneurysm of the

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	rupture) from ICD-10 score linked to operations code according to OPCS-4, linked with all-cause mortality from ONS and ICD-10, only OPCS-4 codes: L18.2, L19.2, L20.2, L21.2			normal adjacent aorta or saccular aneurysm, Life expectancy >2 years <u>Surgical candidate</u> Male or infertile female >21 years old Descending aorta must be clampable distal to the left carotid artery, and distal anastomosis must be performed proximal to the celiac axis Inner aortic diameter of 23-37mm adjacent to aneurysm	patients >18 years old who had undergone an open or endovascular repair of non-ruptured thoracic aortic aneurysm. Patients selected using ICD-9 procedures codes 39.73 and 38.45.	descending thoracic aorta, OSR patients would have been a candidates for ESG due to morphological criteria, Selected on surgeon's opinion of lower risk of ESG

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				Lack of significant thrombus of calcification in landing zones Minimum of 2cm of normal thoracic aorta proximal and distal to aneurysm Aortic taper of no more than 4mm or ability to treat with more than one graft		
Exclusion Criteria	Dissection, Abdominal aneurysms, Non-identified area aneurysm, Ascending			Mycotic aneurysm, Hemodynamically unstable ruptured aneurysm , Major operation (other than planned subclavian to carotid transposition or	NA	Traumatic, Mycotic thoracic aneurysms, Dissections, Aneurysms of the ascending aorta, Aneurysms of the

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	aneurysm, Suprarenal aneurysm, Non-aortic procedures, age <50yrs			bypass) within 30 days, MI or CVA within 6 weeks, Creatinine >2.0mg/dL, Connective tissue disorder, Acute or chronic aortic dissection, Planned occlusion of carotid or celiac arteries, Documented drug abuse within 6 months, Participation in another investigational device or drug study within 1 year		aortic arch, Aortic ulcers
Definition of the short-term	30 days post-procedure	latest occurring of discharge or	latest occurring of discharge or	latest occurring of discharge or 30 days post-intervention	post-procedure hospital stay	30-days post-intervention

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		30 days post-intervention	30 days post-intervention			
PARTICIPANTS						
Country	UK	USA	USA	USA	USA	Italy
Setting	Multi-centre – Hospital episode statistics	Multi-centre – Medicare data	Multi-centre – Medicare data	Multi-centre	Multi-centre	Single-centre
Time frame	2006 to 2011	1998-2007	2003-2007	ESG 1999- 2001, OSR pre 1999 (n=44) & 1999-2001 (n=50)	1998-2007	ESG 2000-2007, OSR 1996-2000
Recruitment	Total 618 ESG 354 OSR 264	Total 13,998 ESG 2433 OSR 11,565	Total 1100 ESG 550 OSR 550	Total 234 ESG 140 OSR 94	Total 8967 ESG 712 OSR 8255	Total 28 ESG 17 OSR 11
Age summary	median (IQR) ESG 73(66,78), OSR 71 (63-76)	mean (95% C.I.)	mean (95% C.I.)	mean(SD) ESG 75.9 (10.4), OSR 68.2 (10.2)	median (IQR) ESG 72 (63- 78), OSR 63 (52, 72)	mean (SD) ESG 66 (10), OSR 61 (13)

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		ESG 75.9 (75.6, 76.1), OSR 73.8 (73.7, 73.9)	ESG 71.1 (70.8, 71.44), OSR 70.7 (70.7, 71.1)			
Proportion male	ESG 232 (65.5%), OSR 137 (51.9%)	% (95% C.I.): ESG 58.7 (56.7, 60.7), OSR 55.4 (54.8, 56.3)	% (95% C.I.): ESG 64.0 (59.9, 68.0), OSR 68.1% (64.2, 72.0)	ESG 90(57%), OSR 48 (51%)	5415 (61.3%) ESG, 437 (65.6%) OSR	ESG 14 (82%), OSR 8 (73%)
Black race	NR	% (95% C.I.): ESG 7.5 (6.4, 8.5), OSR 3.8 (3.4, 4.5) (p=0.001)	% (95% C.I.): ESG 2.7 (1.4, 4.1), OSR 2.7 (1.4, 4.0) (p=0.99)	NR	NR	NR
Charlson comorbidity score	NR	% (95% C.I.): ESG 1.46	n(95% C.I.): ESG 0.5 (0.41,	NR	NR	NR

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		(1.38, 1.54), OSR 0.83 (0.80, 0.86) (p=0.00001)	0.60), OSR 0.44 (0.35, 0.53) (p=0.34)			
Diabetes	NR	% (95% C.I.): ESG 8.00 (6.7, 8.8), OSR 4.10 (3.7, 4.5) (p=0.001)	% (95% C.I.): ESG 4.90 (3.0, 6.7), OSR 2.90 (1.5, 4.3) (p=0.09)	NR	684 (13.2%) ESG, 94 (8.3%) OSR (p<0.001)	NR
Myocardial infarction	NR	% (95% C.I.): ESG 7.00 (6.00, 8.1), OSR 5.10 (4.6, 5.4) (p=0.001)	% (95% C.I.): ESG 2.50 (1.2, 3.8), OSR 3.40 (1.9, 4.9) (p=0.38)	-NR	NR	NR
Congestive heart failure	NR	% (95% C.I.): ESG 10.85%	% (95% C.I.): ESG 3.00 (1.6,	ESG 13 (9%), OSR 9 (10%) (p=1.00)	NR	NR

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		(9.6%- 12.0%), OSR 10.60% (10.1%- 11.2%) (p=0.16)	4.5), OSR 4.70 (2.9, 6.5) (p=0.16)			
Cerebrovascular disease	NR	% (95% C.I.): ESG 5.09% (4.2%-5.9%), OSR 3.20% (2.9%- 3.5%) (p=0.65)	% (95% C.I.): ESG 2.00 (0.5, 2.6), OSR 2.00 (0.8, 3.1) (p=0.65)	NR	NR	4 (23%) ESG, 1 (9%) OSR (p=NS)
COPD	NR	% (95% C.I.): ESG 22.04% (20.3%- 23.6%), OSR	% (95% C.I.): ESG 4.00 (2.3, 5.6), OSR 4.90	ESG 56 (40%), OSR 36 (38%) (p=0.89)	247 (34.7%) ESG, 1483 (17.9%) (P<0.001)	10 (59%) ESG, 6 (54%) OSR (p=NS)

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		10.65% (10.0%-11.2%) (p=0.47)	(3.0, 6.7) (p=0.47)			
Chronic renal failure	NR	% (95% C.I.): ESG 4.9% (4.0%- 5.7%), OSR 0.88% (0.07%-0.10%) (p=0.70)	% (95% C.I.): ESG 0.01 (0, 1.4), OSR 0.50 (0, 1.1) (p=0.70)	NR	90 (12.6%) ESG, 126 (1.5%) OSR (p<0.001)	4 (23%) ESG, 1 (9%) OSR (p=NS)
History of malignancy	NR	% (95% C.I.): ESG 5.71% (4.7%-6.6%), OSR 2.91% (2.6%- 3.2%) (p=0.82)	% (95% C.I.): ESG 1.80 (0.6, 2.9), OSR 1.60 (0.5, 2.6) (p=0.82)	NR	NR	NR

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History of smoking (current or past)	NR	NR	NR	ESG 117 (84%), OSR 77 (82%) (p=0.86)	NR	NR
BMI (kg/m2)	NR	NR	NR	mean (SD): ESG 26.4 (4.7), OSR 26.9 (5) (p=0.44)	NR	NR
Coronary artery disease	NR	NR	NR	ESG 69 (49%), OSR 34 (36%) (p=0.60)	NR	NR
Stroke	NR	NR	NR	ESG 14 (10%), OSR 9 (9%) (p=1.00)	NR	NR
Peripheral arterial disease	NR	NR	NR	ESG 22 (16%), OSR 10 (11%) (p=0.33)	NR	NR
Symptomatic aneurysm	NR	NR	NR	ESG 30 (21%), OSR 36 (38%) (p=0.007)	NR	NR
Other concomitant aneurysm:	NR	NR	NR	ESG 39 (28%), OSR 26 (28%) (p=1.00)	NR	NR
Renal dialysis	NR	NR	NR	ESG 2 (1%), OSR 0 (0%) (p=0.52)	NR	NR

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Paraplegia	NR	NR	NR	ESG 1 (1%), OSR 0 (0%) (p=1.00)	NR	NR
Cancer	NR	NR	NR	ESG 27 (19%), OSR 12 (13%) (p=0.21)	NR	NR
Prior major vascular surgery	NR	NR	NR	ESG 63 (45%), OSR 52 (55%) (p=0.14)	NR	NR
Cardiac comorbidities	NR	NR	NR	NR	265 (37.2%) ESG, 3535 (42.8%) (p<0.001)	NR
Hypertension	NR	NR	NR	NR	NR	16 (94%) ESG, 10 (91%) OSR (p=NS)
Peripheral obstructive disease	NR	NR	NR	NR	NR	5 (29%) ESG, 2 (18%) OSR (p=NS)
Ischemic heart disease	NR	NR	NR	NR	NR	6 (35%) ESG, 5 (45%) OSR (p=NS)

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Cardiac Surgery	NR	NR	NR	NR	NR	(18%) ESG, 0 (0%) OSR (p=NS)
SHORT- TERM OUTCOMES						
All-cause mortality	ESG 21 (5.9%) OSR 4 (1.5%) Adjusted OR: 0.79	ESG (6.1%) 95%CI 5.1%, 7.0%) OSR 7.1% (94%CI 6.7%, 7.6%)	ESG 4.2% (95% C.I. 2.5%-5.8%) OSR 4.5% (95% C.I. 2.8%-6.2%)	ESG 3 (2.1%) OSR 11 (11.7%) Adjusted OR: 0.54	ESG 25 OSR 379	ESG 1 (6%) OSR 2 (20%)
Re-intervention	ESG 21 (5.9%) OSR 4 (1.5%) Adjusted OR: 2.97 (1.09, 8.10)	NR	NR	NR	NR	NR
Paraplegia/ Paraperisis	NR	NR	NR	ESG 4 (3%) OSR 13 (14%)	NR	NR

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Cerebral Vascular Accident	NR	NR	NR	ESG 5 (4%) OSR 5 (4%)	NR	ESG 2 (9%) OSR 1 (12%)
Neurological-unspecified type	NR	NR	NR	NR	ESG 20 (2.8%) OSR 273 (3.3%) Adjusted OR: 0.38	NR
Respiratory failure	NR	NR	NR	ESG 5 (4%) OSR 19 (20%)	NR	NR
Pneumonia	NR	NR	NR	NR	NR	ESG 2 (12%) OSR 3 (27%)
Pulmonary-unspecified type	NR	NR	NR	NR	ESG 17 (2.4%) OSR 462 (5.6%) Adjusted OR: 0.48	NR
Myocardial Infarction	NR	NR	NR	ESG 0 (0%) OSR 1(1%)	NR	ESG 0 (0%) OSR 1 (9%)
Cardiac-unspecified type	NR	NR	NR	NR	ESG 28 (2.9%) OSR 1252 (15.2%)	NR

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					Adjusted OR: 0.24	
Peripheral vascular disease	NR	NR	NR	ESG 20 (14%) OSR 4 (4%)	NR	NR
Renal failure	NR	NR	NR	ESG 2 (1%) OSR 12 (13%)	NR	NR
Wound infection/dehiscence	NR	NR	NR	ESG 5 (4%) OSR 10 (11%)	NR	NR
GI complication	NR	NR	NR	ESG 3 (2%) OSR 6 (6%)	NR	NR
Post-implant syndrome	NR	NR	NR	NR	NR	ESG 4(18%) OSR 0(0%)
LONG-TERM OUTCOMES						
All-cause survival	1 year adjusted OR: 1.1 95%CI [0.70-1.73], p=0.667	NR	NR	NR	NR	NR

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	5 year adjusted HR: 1.45 95%CI [1.08-1.94], p=0.013					
aortic-related survival	5 year adjusted HR: 1.27 95%CI [0.81-1.98], p=0.298	NR	NR	NR	NR	NR
survival from aortic-related re-intervention	5 year adjusted HR: 1.70 95%CI [1.11-2.60], p=0.014	NR	NR	NR	NR	NR

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Appendix 3: Risk of Bias assessment

Risk of Bias Results using the ROBINS-I risk of bias tool

Domain – source of bias	Goodney et al 2011	Goodney et al 2011 (Propensity matched)	Gore TAG	Hughes et al 2014	Piffaretti et al 2007	Von Allmen et al 2014
Confounding	Severe	Moderate	Severe	Severe	Severe	Severe
Selection of participants into the study	Low	Low	Low	Low	Low	Low
Classification of interventions	Low	Low	Low	Low	Low	Low
Deviations from intended interventions	Low	Low	Low	Low	Low	Low
Missing data	Low	Low	Low	Low	Low	Low
Measurement of outcomes	Low	Low	Low	Low	Low	Low
Bias in selection of the reported result	Moderate	Moderate	Moderate	Moderate	Moderate	Moderate
Overall	Severe	Moderate	Severe	Severe	Severe	Severe

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8

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Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency ($e.g., I^2$) for each meta-analysis.	9

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1, appendix 2
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	14, appendix 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	14-19
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	14-19
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	14-19
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA

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DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	19-21
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	21
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	22
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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A Systematic Review of Endovascular Stent Grafting Versus Open Surgical Repair for the Elective Treatment of Arch/Descending Thoracic Aortic Aneurysms

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A Systematic Review of Endovascular Stent Grafting Versus Open Surgical Repair for the Elective Treatment of Arch/Descending Thoracic Aortic Aneurysms

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ABSTRACT

Objective: to review comparisons of the effectiveness of endovascular stent grafting (ESG) against open surgical repair (OSR) for treatment of chronic arch or descending thoracic aortic aneurysms (TAA).

Design Systematic review and meta-analysis

Data sources MEDLINE, EMBASE, CENTRAL, WHO International Clinical Trials Routine data collection, current controlled trials, clinical trials, and the NIHR portfolio were searched from January 1994 through March 2020.

Eligibility criteria for selective studies All identified studies that directly compared ESG and OSR, including RCTs, quasi-randomized and non-randomized controlled trials, comparative cohort studies and case-control studies matched on main outcomes were sought. Study participants had to receive elective treatments for elective arch and/or descending thoracic aortic aneurysms (TAA). Studies were excluded where other thoracic aortic conditions (e.g. rupture or dissection) were reported, unless results for patients receiving elective treatment for arch/descending TAA were reported separately.

Data extraction and synthesis Data was extracted by one reviewer and checked by another. Risk of Bias was assessed using the ROBINS-I tool. Meta-analysis was conducted using a random-effects model. Where meta-analysis not appropriate, results were reported narratively.

Results Five comparative cohort studies met inclusion criteria, reporting a total of 3955 ESG and 21,197 OSR patients. Meta-analysis of unadjusted short-term (30 day) all-cause mortality favoured ESG (OR ratio: 0.75; 95%CI (0.55-0.1.03)). Heterogeneity was identified between larger and smaller studies. Meta-analysis of adjusted short-term all-cause mortality favoured ESG OR 0.71, 95%CI (0.51, 0.98), no heterogeneity. Longer-term (beyond 30 days) survival from all-cause mortality favoured OSR in larger studies and ESG in smaller studies. Freedom from re-intervention in the longer-term

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favoured OSR. Studies reporting short-term non-fatal complications suggest fewer events following ESG.

Conclusions There is limited and increasingly dated evidence on the comparison of ESG and OSR for treatment of arch/descending TAA

POSPERO registration number CRD42017054565

KEY WORDS

Systematic Review, Meta-analysis, Thoracic Aortic Aneurysm, Open Surgical Repair, Endovascular Stent Graft

STRENGTHS AND LIMITATIONS OF THIS STUDY

Strength:

- This is the first systematic review to investigate two-armed studies looking only at elective ESG versus OSR for treatment of chronic arch or descending thoracic aortic aneurysms. Other reviews include dissections, ruptures and treatment for other conditions.
- This systematic review inclusion criteria were broad and sought to include evidence from both randomized and non-randomized study designs.

Limitation:

- Statistical analysis of non-fatal complications was not possible due to the limited and inconsistent reporting of such complications.

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INTRODUCTION

Rationale

A thoracic aortic aneurysm (TAA) is an abnormal expansion of the aorta at any part of its course from the heart to the diaphragm. Aneurysms manifest with acute symptoms or they may be chronic. If untreated, TAA can continue to expand and the annual risk of aneurysm rupture increases from 2% in aneurysms smaller than 5cm in diameter to 7% in aneurysms over 6cm in diameter[1]. The majority of aneurysms (and therefore studies) centre on the ascending or abdominal segments of the aorta, arch and descending aneurysms are relatively under-studied. Therefore, this review focused on chronic aneurysms affecting the arch or descending thoracic aorta.

There are currently two main methods of repair for arch/descending TAA: Open Surgical Repair (OSR) and Endovascular Stent Grafting (ESG). OSR involves replacing the aneurysmal segment of aorta with a surgical graft during open surgery using heart-lung bypass. As the arch and descending aorta supply blood to the brain and spinal cord, OSR is high risk, with reported mortality and paraplegia rates of approximately 5% and 10% [2] respectively. In ESG a stent is placed across the aneurysm, providing a new conduit through which the blood flows. ESG is most often selected for patients who are considered at high risk of death or permanent disability from OSR. However, some patients may not have a suitable anatomy/morphology for ESG procedures [2]. Concerns have also been raised about long-term durability and safety of the implanted devices [3 4] due to complications such as stent migration or endoleaks, with the incidence of late endoleaks (leaks around or through the stent graft that occur more than one year after insertion) reported as being between 2.5% and 27.8% [5]. The need for re-intervention due to technical failures of the stent following ESG is also high (23% at 5 years) [6 7]. With each re-intervention there is an added risk of complication either due to the increased complexity of the procedure or deteriorating health of the patient [2].

One previous systematic review comparing only elective OSR and ESG for descending and arch TAAs has been reported [8]. This systematic review [8] was restricted to data from randomized controlled

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trials in which patients were randomly assigned to ESG and OSR, but no eligible trials were identified when searching relevant data based up to Jan 2016. There are other systematic reviews [9 10] that have included non-randomized studies; however, these reviews also included emergency cases and other thoracic aortic pathologies (e.g. dissection or rupture). As such, they may not represent a true comparison of elective ESG and OSR procedures for descending thoracic aortic aneurysms. Given the lack of RCT evidence, this systematic review includes non-randomized comparative studies of the relative effectiveness of OSR or ESG.

The decision to operate requires a favourable balance between the risks and benefits of that intervention compared to those attached to the expected outcomes of the disease. This balance is in turn influenced by a number of factors. Aneurysm size correlates to the risk of death (if left untreated) and so patients are usually referred once the aneurysm crosses a chosen threshold. At that point, patient fitness (*and life expectancy*), the reality or possibility of an intrinsic weakness affecting the whole aorta, the technical feasibility of the proposed intervention and patient wishes, come into play. This systematic review takes a step back from these issues and examines only the outcomes for patients in whom a decision to intervene has been made. This review is an essential starting point for the analysis of effective treatments for thoracic aortic aneurysms.

Objectives

The aim of this systematic review was to assess the effectiveness of endovascular stent grafting (ESG) compared to open surgical repair (OSR) as a treatment for chronic arch or descending TAA.

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METHODS

Protocol and registration

The review protocol was registered in the PROSPERO database (CRD42017054565)[11].

PUBLIC PATIENT INVOLVEMENT

There was no patient and public involvement in the whole process of conducting this review.

Eligibility criteria

Study designs

All identified studies that directly compared ESG and OSR, including RCTs, quasi-randomized and non-randomized controlled trials, comparative cohort studies and case-control studies matched on main outcomes were sought [12].

Non-randomized studies that controlled for selection bias were included – irrespective of study size. Acceptable methods for addressing selection bias included matching patients at baseline for important covariates, matching patients using propensity scores or adjustment for important covariates during analysis. Non-randomized studies that did not use these methods but included larger sample sizes (more than 250 patients) were also included as they have the potential to reduce sampling variation relative to smaller unmatched studies[13].

Non-English language papers were noted and, where possible, data were extracted from the abstract. No translation of full text articles was possible.

Study Participants

Study participants had to receive elective treatments for arch and/ or descending thoracic aortic aneurysms. Where emergency interventions or other thoracic aortic conditions (such as intramural haematoma, rupture, and dissection) were reported, studies were excluded, unless results for patients receiving elective treatment for arch/descending TAA were reported separately. Studies

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that reported secondary (re-intervention) procedures for arch/descending TAAs were included only if the original, primary procedure for each participant could be identified.

Study interventions

Open Surgical Repair was defined as replacement of the aneurysmal aorta with a prosthetic conduit via a sternotomy or thoracotomy with circulatory support. Endovascular stent grafting was defined as transluminal introduction of a stent-graft under X-ray guidance.

Outcome measures

This review assessed short-term and long-term clinical effectiveness. Short-term was defined as either within 30-days of the intervention and/or time until discharge following the intervention. Since most studies did not report long-term outcomes conditional on surviving the short-term period, long-term analyses covered the time from intervention until the end of follow-up. Primary outcomes were all-cause mortality and aneurysm-related mortality. Secondary outcomes were re-interventions and non-fatal complications. The latter were sub-classified as vascular, neurological, cardiac, respiratory and 'other' complications.

Information sources

MEDLINE, EMBASE, CENTRAL, WHO International Clinical Trials Routine data collection, Current Controlled Trials, Clinical Trials, and the UK National Institute for Health Research portfolio were searched. The search strategy (Appendix 1) was adapted as appropriate for each database (for example to allow for variations in controlled vocabulary terms and syntax [14]). Electronic searches identified publications between 1994, date of first reported case of ESG for descending and arch TAA[15], and March 2020.

Websites of regulatory bodies and Health Technology Assessment agencies were screened for unpublished relevant reports. A citation search of all included studies was conducted via Web of

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Science. Studies from systematic reviews that satisfied the inclusion criteria of this review were screened for eligibility.

Study selection

All titles and abstracts were screened by one reviewer (AM). A second investigator (JG) reviewed 10% of all titles and abstracts classed as eligible or not eligible and all titles and abstracts classified as unclear by the first reviewer. The full text of selected studies was assessed by one investigator (AM) and classified as relevant, not relevant or unclear according to the inclusion criteria. A second investigator (JG) reviewed 10% of all papers, as well as all studies deemed unclear. Disagreements at either stage were resolved by discussion and where necessary a third reviewer (PM) arbitrated. Where required, a clinical expert (PS) was asked to screen papers.

Data collection process

A data extraction form was developed based on The Cochrane Collaboration data collection form [16]. Data were extracted by one reviewer (AM) and checked by a second reviewer (JG) for accuracy. Disagreements were resolved by discussion or arbitration by a third reviewer (PM).

Data items

Reported short-term and longer-term all-cause mortality, as well as short-term and longer-term aneurysm-related mortality were collected. All reported complications were recorded and grouped according to expert clinical guidance as described above as neurological, cardiac, respiratory or 'other' complications, with endoleaks following ESG recorded separately.

Risk of bias in individual studies

Risk of bias tools used were specific to the study design. As described later only non-randomized study designs were identified. Therefore, the ROBINS-I tool [17] was used. The ROBINS-I tool classifies the risk of bias in seven domains (*confounding; selection of participants; classification of*

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interventions; deviations from intended interventions; missing data; measurement of outcomes; selection of the reported results) and overall as either critical, severe, moderate, low, or unclear.

Synthesis of results

Point estimates and precision of the incidence of short and long-term events and unadjusted and adjusted comparisons (odds ratios (OR) hazard ratios (HR)) were recorded. Meta-analysis of short-term event-comparisons was based on the OR with 95% confidence interval (CI) and conducted using RevMan [18]. An overall treatment effect was calculated for the log (OR) of each outcome using a (normal) random effects model. Where comparisons could not be made, data were tabulated and reported narratively.

Heterogeneity between studies was assessed according to Cochrane guidelines, on clinical criteria, by visual inspection of plots of the data, from the chi-square test for heterogeneity, and the I-squared statistic. An I^2 statistic of 25-50% indicated low heterogeneity; 50-75% indicated moderate heterogeneity, and over 75% indicating significant heterogeneity.

RESULTS

Study selection

The search identified 5163 unique studies. Of these, 5078 studies were not eligible. Ineligibility was primarily because of irrelevant anatomy (ascending, abdominal, or peripheral aneurysms), irrelevant pathology (acute dissection or other acute aortic syndromes), or because the study pooled data from a mixture of anatomical and pathological situations such that data on chronic arch and DTA aneurysms could not be extracted with confidence. Thus 85 papers underwent full text screening. 75 papers were rejected at this stage leaving 10 papers to be included in the full review. Six papers reported data from the Gore TAG Trial [19-24], leaving five unique comparative studies that fulfilled the inclusion criteria. Further details of the selection process are shown in the PRISMA diagram (Figure 1).

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One study, Goodney et al 2011, reported both an unadjusted analysis for the unadjusted cohort and an analysis of a subgroup of propensity score matched patients. This review reports information from both cohorts separately but does not include both in any analysis to avoid double-counting. Brief summaries of the identified studies are given in Table 1, with detail provided in Appendix 2. Overall, the five studies included data on 3955 ESG and 21,197 OSR patients and all studies were conducted in Europe or the USA. Two studies had non-overlapping or only partially overlapping recruitment periods between the ESG and OSR intervention groups (Gore Tag and Piffaretti et al 2007[19 25]). There were clear differences in patient characteristics between the two treatment groups. ESG patients were consistently older, with average age differences ranging from 2 to 9 years in all but one analysis (Goodney et al 2011 the propensity matched cohort). Similarly, in most studies the proportion of men was greater in the ESGgroup.

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Table 1: Study characteristics

Study	**Von Allmen et al 2014**	**Goodney et al 2011 (2 analyses*)**	**Gore TAG**	**Hughes et al 2014**	**Piffaretti et al 2007**
METHODS					
Study Design	Comparative cohort	Comparative cohort	Comparative cohort	Comparative cohort	Comparative cohort
Method of matching	None	1. None 2. propensity matched	ESG patients judged eligible for OSR	None	OSR patients judged eligible for ESG
PARTICIPANTS					
Country	UK	USA	USA	USA	Italy
Setting	Multi-centre – Hospital episode statistics	Multi-centre – Medicare data	Multi-centre	Multi-centre	Single-centre
Time frame	2006 to 2011	1. 1998-2007 2. 2003-2007	ESG 1999-2001, OSR pre 1999 (n=44) & 1999-2001 (n=50)	1998-2007	ESG 2000-2007, OSR 1996-2000
Recruitment	Total 618 ESG 354 OSR 264 OSR	1. Total 13,998 (ESG 2433, OSR 11,565)	Total 234 ESG 140 OSR 94	Total 8967 ESG 712 OSR 8255	Total 28 ESG 17 OSR 11

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		2. Total 1100 (ESG 550, OSR 550)			
Age summary	median (IQR) ESG 73(66,78), OSR 71 (63-76)	mean (95% C.I.) 1. ESG 75.9 (75.6, 76.1), OSR 73.8 (73.7, 73.9) 2. ESG 71.1 (70.8, 71.4), OSR 70.7 (70.7, 71.1)	mean(SD) ESG 75.9 (10.4), OSR 68.2 (10.2)	median (IQR) ESG 72 (63-78), OSR 63 (52, 72)	mean (SD) ESG 66 (10), OSR 61 (13)
Proportion male	ESG 232 (65.5%), OSR 137 (51.9%)	% (95% C.I.): 1. ESG 58.7 (56.7, 60.7), OSR 55.4 (54.8, 56.3) 2. ESG 64.0 (59.9, 68.0), OSR 68.1% (64.2, 72.0)	ESG 90(57%), OSR 48 (51%)	5415 (61.3%) ESG, 437 (65.6%) OSR	ESG 14 (82%), OSR 8 (73%)
OUTCOMES					
Short-term	Death from any cause	Death from any cause	Death from any cause	Death from any cause	Death from any cause

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	Re-intervention All-cause mortality (unadjusted		Complications	Complications	Complications
Long-term	Time to death from any cause Time to aortic-related death Time to aortic-related re-intervention	Time to death from any cause	Time to death from any cause Time to aneurysm-related death aneurysm-related mortality Time to re-intervention Time to major adverse event	Not reported	Time to death from any cause
* 1. Unmatched cohort, 2. Propensity matched cohort					

Risk of bias within studies

The propensity-matched analysis of Goodney et al (2011) [26] was judged to be of overall moderate risk of bias and four studies were at severe risk of bias due to the potential for confounding (Appendix 3).

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Two large cohorts derived from UK and USA administrative databases (Von Allmen et al 2014 and Hughes et al 2014) [27 28] did not attempt to address selection bias at the design stage, but did adjust for some confounders in the analysis. A third study (Goodney et al 2011), using USA administrative data, included both unadjusted population and analysis of a subset of the data defined by propensity score matching. Propensity score methods rely on all important confounders being available and included[29]. As this study did not include all confounders considered important, such as aneurysm size, this study was judged to have moderate risk of bias due to confounding.

One study (Gore TAG)[19] included only ESG patients who would also have been eligible for OSR and the another study (Piffaretti et al 2007) [25] included only OSR patients who would be suitable for ESG, both according to clinician judgement. Since suitability was decided retrospectively, such matching may not be robust.

A further potential bias with each study was selective reporting. Although the measurements and analyses of the outcomes were clearly defined within each study, no evidence of a pre-study analysis plan was identified.

Synthesis of results

Short-term death from any cause (30-day or in hospital)

Unadjusted Mortality:

All five studies [19 25-28] reported unadjusted 30 day or in-hospital mortality (Figure 2). Overall, the odds of short-term mortality were lower in the ESG group than the OSR group (OR: 0.75; 95 %CI. (0.55-1.03)). Moderate heterogeneity was present in the overall analysis (I^2= 40%), mainly due to differences between the two smaller studies [19 25] and the three larger studies [26-28], with a large I^2 for subgroup differences between the larger studies subgroup and the smaller studies subgroup of 84.5%.

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For the three larger studies[26-28] (3798 ESG and 21,092 OSR patients), the treatment effect for death was smaller (OR: 0.84 95%CI (0.72, 0.98)) than that for the two smaller studies [19 25](157 ESG and 105 OSR patients, combined OR: 0.18 95%CI (0.06, 0.59)). There was no evidence of heterogeneity in results between these subgroups (I^2=0).

Sensitivity analysis

Hughes et al 2014 was the only study to include aortic arch aneurysms. Arch aneurysms tend to be more complex procedures and are mostly conducted using OSR. A sensitivity analysis excluding this study was therefore conducted (Figure 3).

Overall, the odds of short-term mortality were lower in the ESG group than the OSR group (OR: 0.73; 95 %CI. (0.45-1.18)). Moderate heterogeneity was present in the overall analysis (I^2= 42%), mainly due to differences between the two smaller studies and the two larger studies, with a large I^2 for subgroup differences between the larger studies subgroup and the smaller studies subgroup of 65.1%.

For the two larger studies (3086 ESG and 12837 OSR patients), the treatment effect for death was smaller (OR: 0.85 95%CI (0.72, 1.00)) than that for the two smaller studies (157 ESG and 105 OSR patients, combined OR: 0.20 95%CI (0.04, 1.06)). There was no evidence of heterogeneity in results within the large subgroup (I^2= 0) and evidence of low heterogeneity withing the small studies subgroup (I^2=26%).

Adjusted Mortality:

The three larger studies [26-28] included adjusted analysis of short-term mortality (Figure 4). Goodney et al 2011 used stratified propensity matching. Hughes et al 2014 [27] adjusted for preoperative comorbidities, independent predictors of mortality and post-operative complications and the Von Allmen 2014 study adjusted for age and sex. The OR for ESG, relative to OSR, for these three studies

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combined was 0.71, 95%CI (0.51, 0.98). No evidence of heterogeneity was identified in this meta-analysis (I^2=0%)

All-cause mortality (following hospital discharge or after 30 days)

Meta-analysis of the longer-term all-cause mortality was not possible as follow-up varied between the four studies that reported this outcome (Gore Tag, Goodney et al 2011, Piffaretti et al 2007 and Von Allmen 2014) [19 25 26 28] . The results in the larger studies (Goodney et al 2011, Von Allmen 2014)[26 28] differed from those in the smaller studies (Gore Tag and Piffaretti et al 2007)[25 30]. We note that the analyses included the short-term deaths as well as those occurring beyond hospital discharge or after 30 days.

Von Allmen et al 2014 reported time to death from any cause up to five years, throughout which OSR had higher overall survival. Adjusting for age and sex, ESG had higher odds of death at one year (OR: 1.10 95%CI (0.70, 1.73); p=0.667). Additionally, the hazard ratio for ESG relative to OSR up to five years, adjusting for age and sex, was HR=1.45 (95%CI [1.08, 1.94], p=0.013).

Goodney et al 2011 also reported time to death from any cause up to five years. Adjusting for age, sex, race, era of procedure and Charlson comorbidity score, five-year survival for OSR in the unmatched cohort was higher than for ESG (89% versus 79%, log-rank p<0.0001). In the propensity-matched cohort, five-year life-table analysis reported higher survival in the OSR group (81% 95%CI (77%-85%)) than for the ESG group (73% 95%CI (68%, 76%)) (log-rank p=0.007).

In the Gore TAG study, Time to death from any cause over five years did not differ between the ESG and OSR groups, with survival probability at five years in both groups of 63% (log rank P value= 0.625). Piffaretti et al 2007 reported survival overall in the ESG group up to five years and the OSR group up to 13 years. For the five years in which survival is reported in both groups, ESG had a higher

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survival compared to OSR although this was based on only 4 deaths, and the follow-up periods differed substantially (log rank p>0.05).

Aneurysm/Aortic-related mortality

Meta-analysis of Aneurysm/aortic-related mortality could not be conducted. No studies reported aneurysm-related mortality during hospitalization or up to 30 days separately to the long-term. Von Allmen et al 2014 defined aortic-related mortality as any aortic-related death occurring within five years of the index operation according to ICD-10 codes referring to aorta, or cause of deaths closely related to complications of surgery (e.g. paraplegia). Time to aortic-related death did not differ between OSR group than the ESG group during the five-year follow-up (log-rank p=0.270). This study[28] also reported adjusted (for age and sex) five year HR of 1.27 (0.81 to 1.98; P =0.298) for ESG patients relative to OSR patients.

Gore TAG[19] defined aneurysm related mortality as deaths that occurred in hospital or within 30 days of initial procedure or any reintervention or a death due to the aneurysm or the treatment device and reported up to five years, with longer time to aneurysm-related death in the ESG group over this period (log-rank p= 0.012).

Short-term re-interventions

One study reported aortic-related re-interventions in 21 (5.9%) ESG patients and 4 (1.5%) OSR patients, an adjusted OR of 2.97 (95%CI [1.09, 8.10], p=0.033) for ESG versus OSR (Von Allmen et al 2014[28]).

Long-term re-interventions

Two studies reported this outcome (Von Allmen et al 2014[28] and Gore TAG [19]). Von Allmen et al 2014[28] reported time to aortic-related re-interventions up to five years. Time to re-intervention was

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longer in the OSR group compared to the ESG group (log-rank p=0.007), with adjusted (for age and sex) hazard ratio at five years of 1·70 (95%CI [1·11 to 2·60], p =0·014). In Gore TAG[19], time to re-intervention over five years was longer in the ESG group (log-rank P=0.011). We note that these analyses included short-term re-interventions.

Complications

Table 2 summarises the short-term complications reported in all five studies[19 25-28]. Most studies reported a small number of neurological, pulmonary, cardiac and other complications. However, one larger study (Hughes et al 2014)[27] reported lower odds for ESG patients for neurological complications, OR of 0.48 (95%CI (0.26, 0.86), p=0.015), pulmonary complications, OR: 0.38 (95%CI (0.21, 0.67), p=0.001) and cardiac complications, OR: 0.24 (95%CI (0.15, 0.37), p<0.001). The Gore TAG study[19] reported endoleak rates for ESG patients of 8.5%.

Table 2: Short-term complications

Study	**Gore TAG**		**Piffaretti et al 2007**		**Hughes et al 2011**	
Arm (participants)	**ESG (n=140)**	**OSR (n=94)**	**ESG (n=17)**	**OSR (n=11)**	**ESG (n=712)**	**OSR (n=8255)**
NEUROLOGICAL						
Paraplegia	NR	NR	NR	NR	NR	NR
Paraplegia/ Paraperisis	4 (3%)	13 (14%)	NR	NR	NR	NR
Cerebral Vascular Accident	5 (4%)	5 (4%)	2 (9%)	1 (12%)	NR	NR
Neurological-unspecified type	NR	NR	NR	NR	20 (2.8%)	273 (3.3%)
RESPIRATORY						

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Respiratory failure	5 (4%)	19 (20%)	NR	NR	NR	NR
Pneumonia	NR	NR	2 (12%)	3 (27%)	NR	NR
Pulmonary-unspecified type	NR	NR	NR	NR	17 (2.4%)	462 (5.6%)
CARDIAC						
Myocardial Infarction	0 (0%)	1(1%)	0 (0%)	1 (9%)	NR	NR
Cardiac-unspecified type	NR	NR	NR	NR	28 (2.9%)	1252 (15.2%)
ENDOLEAKS						
Endoleaks	12 (8.5%)	NR	NR	NR	NR	NR
OTHER						
Peripheral vascular disease	20 (14%)	4 (4%)	NR	NR	NR	NR
Renal failure	2 (1%)	12 (13%)	NR	NR	NR	NR
Wound infection/dehiscence	5 (4%)	10 (11%)	NR	NR	NR	NR
GI complication	3 (2%)	6 (6%)	NR	NR	NR	NR
Post-implant syndrome	NR	NR	4(18%)	0(0%)	NR	NR
Note: Goodney et al 2011 and Von Allmen et al 2014 did not report short-term complications						

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DISCUSSION

This is the first systematic review that reports results from non-randomized comparative studies regarding the effectiveness of elective ESG and OSR for arch and or descending TAA. Previous reviews have either included single arm studies only[31] or were comparative reviews that included non-elective procedures[9 32], or mixed pathologies such as dissections[10 33]. The results of these

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reviews may not be applicable to patients undergoing elective procedures for descending or arch TAA pathology.

In this review only five comparative cohort studies were identified. The lack of RCTs comparing ESG and OSR for elective arch/descending thoracic aortic aneurysms may reflect the lack of equipoise from practitioners and patients. Well-designed non-randomized trials can demonstrate a high degree of concordance with the results of well-designed RCTs, provided that all confounders are adequately accommodated in the design and/or analysis[34]. However, the studies identified in this review had moderate to serious risk of bias, mainly due to potential for confounding.

Although, complications in the short-term also appear to favour ESG, evidence regarding short-term complications is either lacking or inconsistently reported. All five studies[19 25-28] suggested that the less invasive nature of ESG confers lower short-term mortality and fewer short-term non-fatal complications than OSR. The two small unadjusted studies[19 25] had a much larger effect than larger studies[26-28]. These larger studies used more robust methods for addressing confounding; this may indicate the presence of publication bias where only those smaller studies with larger effect sizes have been published.

Longer-term results are difficult to interpret since most data were not presented in a way to separate early risk from longer term survival. The two smaller studies[19 25] reported continuing survival benefit for ESG patients in the long-term but are limited by small numbers, different recruitment periods, and failure to exclude early risk. The survival advantage associated with ESG in the Gore TAG study disappears by year five, which suggests that the hazards are lower for OSR patients that survive the initial procedure. Conversely, the two larger studies (Von Allmen et al 2014 and Goodney et al 2011[26 28]) report that the short-term survival advantage of ESG were reversed by twelve months, and that the OSR group hadan increasing survival advantage thereafter. Similarly, time to re-intervention favoured ESG in the smaller Gore TAG study, whilst time to aortic-related re-interventions favoured OSR in the larger study Von Allmen et al 2014[28].

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The evidence regarding mortality is inconsistent across the studies with more recent, larger studies[26-28] reporting a more modest survival benefit for ESG in the short-term and a survival benefit for OSR in the long-term. The two[19 25], smaller studies suggest a large reduction in the risk of mortality and short term complications in the short run and a survival benefit in the long-run for ESG.

Inconsistency of results across studies may be due to heterogeneity of the study populations, both within and across studies. Differences in recruitment periods may reflect changes in technology and practice in both groups. The larger studies [26-28] were based on patients recruited up to twenty years later than the smaller studies. Outcomes may have improved differentially with OSR patients receiving proportionally better outcomes due to general improvements in surgical practice and patient selection over time[35]. A potential reason for the reported long-term survival benefit of OSR in more recent, larger studies may be differences in patient risk in each intervention group. Open surgical repair is advocated for younger, lower-risk patients, and EVAR is preferred for older, higher-risk patients[36]. This selection bias may not have been adequately accommodated in adjusted analyses.

The three large studies[26-28] were based on administrative health records with case ascertainment that depends on accurate coding, which can vary across and within providers[37]. Furthermore, one of the small studies, Piffaretti et al 2007[25], recruited patients from one centre which may have compromised external validity as they may for example, depend on the expertise within a centre and the nature of the patients that centre treats[30 38]. Additionally, the Gore TAG study[19] utilised a more restricted inclusion criteria for ESG patients compared to those who received OSR, potentially introducing heterogeneity between the two groups.

The headline conclusion of this review is that whilst there is a degree of variability, OSR is better for long-term survival and avoidance of re-intervention, and that ESG is better for early outcomes. This conclusion is based on non-randomized comparisons in which the influence of important

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confounders for choice of intervention (ESG or OSR) has not been adjusted for. Deeper exploration is required, but randomized data may be difficult to acquire as there are well-established guidelines about which patients are favoured for OSR and which for ESG. Thus, to compare ESG and OSR it is first necessary to establish an understanding of the reasons driving clinical management, for example by a structured consensus analysis such as a Delphi study. Thereafter, unless there is clinical equipoise about which patients are best served by OSR or ESG, the best quality evidence to establish the relative merits of ESG and OSR for arch/ DTA aneurysms is likely to be a systematic, prospective, observational, comparative study, using individual patient data.

Limitations

The findings in this review are limited by the number and potential for confounding of the five studies[19 25-28] identified, so that definitive conclusions are not possible. The adjustment for confounders is likely to be only partial and some bias in the estimates of differences in mortality and complications may persist, despite attempts to reduce bias in the larger studies identified. Statistical analysis of non-fatal complications was not possible due to limited and inconsistent reporting.

CONCLUSIONS

This systematic review is the first to consider evidence from non-randomized studies that directly compare ESG and OSR for the treatment of elective arch/descending thoracic aortic aneurysms. Other reviews conducted in this area have been complicated by the inclusion of treatments for emergency cases or other thoracic aortic pathologies (e.g. thoracic aortic ulcer, or dissection).

There is limited and increasingly dated evidence addressing the comparison of ESG and OSR in the management of elective arch and or descending thoracic aortic aneurysms. Further, high quality, evidence is required. Although large RCTs may not be possible in this relatively rare condition, future studies should use strong methodological study design in order to control for potential confounders

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and reduce potential biases. Additionally, given the current conflicting evidence, it is important that comparisons of OSR and ESG are conducted in both the short-term and the long-term.

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CONFLICT OF INTEREST

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DATA SHARING AGREEMENT

Data were extracted from the studies included in this Systematic Review and referenced below.

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Figure 1: PRISMA Diagram

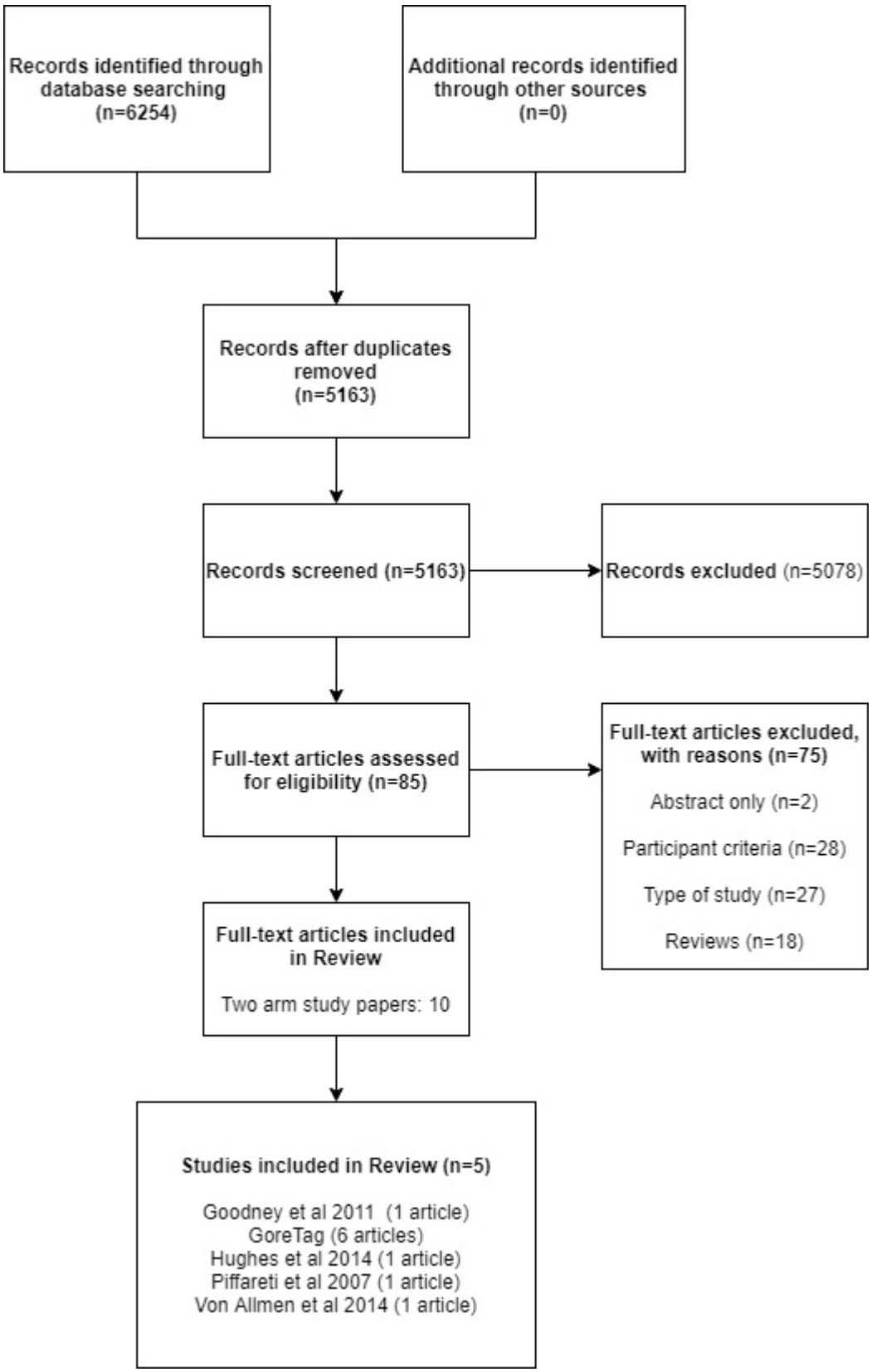
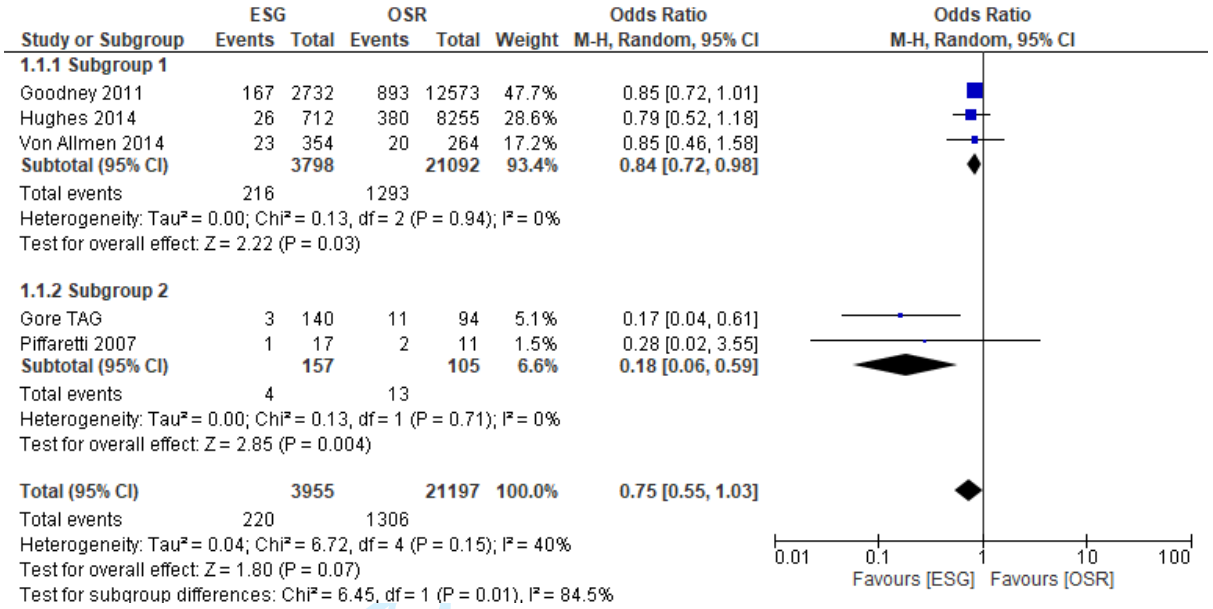
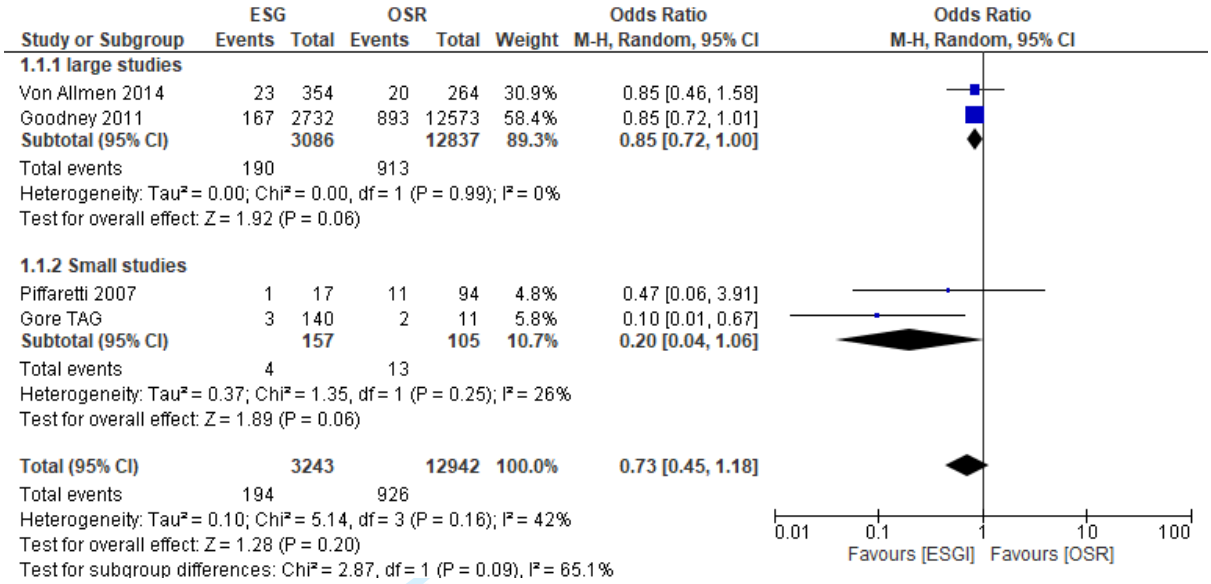


Figure 2: Forest plot of unadjusted short-term all-cause mortality



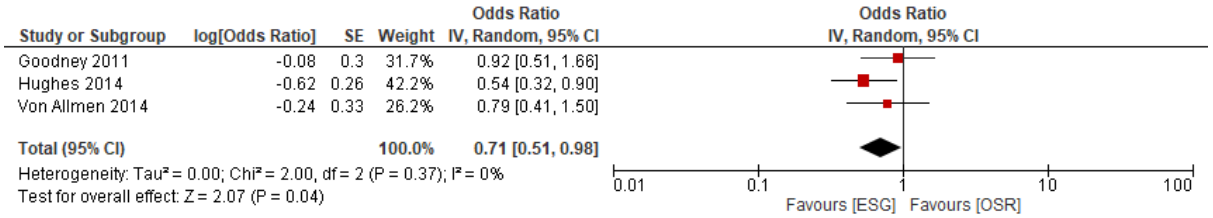
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Figure 3: Forest plot of unadjusted short-term all-cause mortality. Sensitivity Analysis.



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Figure 4: Forest plot of adjusted short-term all-cause mortality



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Appendix 1: Search Strategy

Medline/Central/HTA

1. (TEVAR or EVAR).ti,ab.
2. (endovascular adj1 (stent* or repair* or treat* or surg* or (aneurysm? adj1 repair*))).ti,ab.
3. (open adj2 (surg* or repair*)).ti,ab.
4. endoprosthe*.mp.
5. *Aortic Aneurysm, Thoracic/su [Surgery]
6. *Aortic Thoracic/su [Surgery]
7. Thoracic Surgical Procedures/
8. or/1-7
9. exp Aortic Aneurysm, Thoracic/
10. (thorac* adj2 aneurysm*).ti,ab.
11. (descending adj1 aort* adj1 aneurysm*).ti,ab
12. TAA*.ti,ab.
13. (arch adj3 aneurysm*).ti.ab.
14. or/9-13
15. morbid*.mp.
16. mortalit*.mp.
17. outcome?.mp.
18. complication*.mp.
19. (rupture? or endoleak?).mp.
20. device migration.mp.
21. (ae or co or de).fs

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22. (safe or safety or side effect* or undesirable effect* or treatment emergent or tolerability or toxicity or adrs or (adverse adj2 (effect or effects or reaction or reactions or event or events to outcome or outcomes))).ti,ab.
23. ((technical or surg*) adj2 (advance* or improve*)).mp.
24. (quality adj2 life).mp.
25. Mortality/
26. Mortbidity/
27. Postoperative complications/
28. Intraoperative complications/
29. Aortic Aneurysm, Thoracic/co
30. Conversion to Open Surgery/
31. “Quality of Life”
32. or/15-31
33. 8 and 14 and 32
34. (“clinical trial” or “clinical trial, phase I” or “clinical trial, phase ii” or clinical trial phase iii or clinical trial, phase iv or controlled clinical trial or “multicentre study” or “randomized controlled trial”).pt. or double-blind method/ or clinical trials, phase iii as topic/ or clinical trials, phase iv as topic/ or controlled clinical trials as topic/ or randomized controlled trials as topic/ or early termination of clincal trials as topic/ or multicenter studies as topic/ or ((randomi?ed adj7 trial*) or (controlled adj3 trial*) or (clinical adj2 trial*) or ((single or doubl* or tripl* or treb*) and (blind* or mask*))).ti,ab,kw. or (“4 arm” or “four arm”).ti,ab,kw.
35. cohort studies/ or longitudinal studies/ or follow-up studies/ or prospective studies/ or retrospective studies/ or cohort.ti,ab. or longitudinal.ti,ab. or prospective.ti,ab. or retrospective.ti,ab. or quasi-experiment.ti,ab.

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36. Clinical Trial.pt. or random*.tw. or Clinical Trials/ or clinical trial.tw. or Follow-up Studies/ or prospectiv*.tw. or control*.tw.
37. or/34-36
38. 33 and 37

EMBASE

1. (TEVAR or EVAR).ti,ab.
2. (endovascular adj1 (stent* or repair* or treat* or surg* or (aneurysm? adj1 repair*))).ti,ab.
3. (open adj2 (surg* or repair*)).ti,ab.
4. endoprosthe*.mp.
5. Thoracic Aorta Aneruysm/su [Surgery]
6. Thoracoabdominal aorta aneurysm/su [Surgery]
7. Thoracic Aorta/su [Surgery]
8. exp Thoracic aorta surgery/
9. or/1-8
10. exp Thoracic Aorta Aneurysm/
11. exp Thoracoabdominal aorta aneurysm/
12. Thoracic Aorta/
13. (thorac* adj2 aneurysm*).ti,ab.
14. (descending adj1 aort* adj1 aneurysm*).ti,ab.
15. TAA*.ti,ab.
16. (arch adj3 aneurysm*).ti,ab.
17. or/10-16
18. (outcome? or mortalit*).mp.
19. complication*.mp.
20. ((technical or surg*) adj2 (advance* or improve*)).mp.

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21. (quality adj2 life).mp.
22. Mortality/
23. Postoperative complication/
24. peroperative complication/
25. Thoracic Aorta Aneurysm/co
26. “Quality of Life”/
27. or/18-26
28. (Randomised Controlled Trial or Pragmatic Clinical Trial).pt or exp Randomized Controlled Trials as Topic/ or “Randomized Controlled Trial (topic)”/ or Randomized Controlled Trial/ or Randomization/ or Random Allocation/ or Double-Blind Method/ or Double Blind Procedure/ or Double-Blind Studies/ or Single-Blind Method/ or Single Blind Procedure/ or Single-Blind Studies/ or Placebos/ or Placebo/ or (random* or sham or placebo*).ti,ab,hw,kf,kw. or ((singl* or doubl*) adj (blind* or dumm* or mask*)).ti,ab,hw,,kw. or ((tripl* or trebl*) adj (blind* or dumm* or mask*)).ti,ab,hw,,kw.
29. random:.tw. or placebo:.mp. or double-blind:.mp.
30. controlled clinical trial/ or clinical trial/ or exp controlled study/ or randomized controlled trial/ or exp quasi experimental study/
31. or/28-30
32. 9 and 17 and 27 and 31
33. limit 32 to (human and yr=”1994-Current”)
34. exp cohort analysis/ or exp longitudinal study/ or exp prospective study/ or exp follow up/ or cohort*.tw.
35. limit 34 to (human and yr=”1994-Current”)
36. 9 and 17 and 27 and 35
37. 36 not 33
38. 33 or 37

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Appendix 2: Data extraction table

Study	**Von Allmen et al 2014**	**Goodney et al 2011** (2 analyses*)	**Gore TAG**	**Hughes et al 2014**	**Piffaretti et al 2007**
METHODS					
Study Design	Comparative cohort	Comparative cohort	Comparative cohort	comparative cohort	Comparative cohort
Method of matching	None	1. None 2. propensity matched	ESG patients judged eligible for OSR	None	OSR patients judged eligible for ESG
Inclusion Criteria	Hospital diagnosis code 171.2 (TAA without mention of rupture) from ICD-10 score linked to	Patients diagnosed with a thoracic aneurysm according to ICD9 diagnosis code:	Fusiform descending thoracic aortic aneurysm at least twice the size of the normal adjacent aorta or saccular	Nationwide inpatient Sample database of patients >18 years old who had	Elective, Chronic fusiform or saccular aneurysm of the descending thoracic aorta, OSR patients

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	operations code according to OPCS-4, linked with all-cause mortality from ONS and ICD-10, only OPCS-4 codes: L18.2, L19.2, L20.2, L21.2	Intact thoracic aneurysms 441.2 Patients had undergone an open or TEVAR procedure according to ICD9 procedure codes: Open 38.35, 38.45 TEVAR 39.73, 39.79	aneurysm, Life expectancy >2 years Surgical candidate Male or infertile female >21 years old Descending aorta must be clampable distal to the left carotid artery, and distal anastomosis must be performed proximal to the celiac axis Inner aortic diameter of 23-37mm adjacent to aneurysm Lack of significant thrombus of calcification in landing zones	undergone an open or endovascular repair of non-ruptured thoracic aortic aneurysm. Patients selected using ICD-9 procedures codes 39.73 and 38.45.	would have been a candidates for ESG due to morphological criteria, Selected on surgeon's opinion of lower risk of ESG

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			Minimum of 2cm of normal thoracic aorta proximal and distal to aneurysm Aortic taper of no more than 4mm or ability to treat with more than one graft		
Exclusion Criteria	Dissection, Abdominal aneurysms, Non-identified area aneurysm, Ascending aneurysm, Suprarenal aneurysm, Non-	Patients with ICD 9 diagnosis codes indicating ascending thoracic aneurysms or cardiopulmonary bypass occurring with circulatory arrest. Thoracoabdominal aneurysm, thoracic aortic	Mycotic aneurysm, Hemodynamically unstable ruptured aneurysm , Major operation (other than planned subclavian to carotid transposition or bypass) within 30 days, MI or CVA within 6 weeks, Creatinine >2.0mg/dL, Connective tissue	NA	Traumatic, Mycotic thoracic aneurysms, Dissections, Aneurysms of the ascending aorta, Aneurysms of the aortic arch, Aortic ulcers

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	aortic procedures, age <50yrs	dissection, other aortic pathology. IC9 procedure codes indicating debranching or other procedures to extend endovascular landing zones (39.24, 39.25,	disorder, Acute or chronic aortic dissection, Planned occlusion of carotid or celiac arteries, Documented drug abuse within 6 months, Participation in another investigational device or drug study within 1 year		
Definition of the short-term	30 days post-procedure	latest occurring of discharge or 30 days post-intervention	latest occurring of discharge or 30 days post-intervention	post-procedure hospital stay	30-days post-intervention
PARTICIPANTS					
Country	UK	USA	USA	USA	Italy

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Setting	Multi-centre – Hospital episode statistics	Multi-centre – Medicare data	Multi-centre	Multi-centre	Single-centre
Time frame	2006 to 2011	1. 1998-2007 2. 2003-2007	ESG 1999- 2001, OSR pre 1999 (n=44) & 1999-2001 (n=50)	1998-2007	ESG 2000-2007, OSR 1996-2000
Recruitment	Total 618 ESG 354 OSR 264	1. Total 13,998 (ESG 2433, OSR 11,565) 2. Total 1100 (ESG 550, OSR 550)	Total 234 ESG 140 OSR 94	Total 8967 ESG 712 OSR 8255	Total 28 ESG 17 OSR 11
Age summary	median (IQR) ESG 73(66,78), OSR 71 (63-76)	mean (95% C.I.) 1. ESG 75.9 (75.6, 76.1), OSR 73.8 (73.7, 73.9) 2. ESG 71.1 (70.8, 71.44), OSR 70.7 (70.7, 71.1)	mean(SD) ESG 75.9 (10.4), OSR 68.2 (10.2)	median (IQR) ESG 72 (63- 78), OSR 63 (52, 72)	mean (SD) ESG 66 (10), OSR 61 (13)

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Proportion male	ESG 232 (65.5%), OSR 137 (51.9%)	% (95% C.I.): 1. ESG 58.7 (56.7, 60.7), OSR 55.4 (54.8, 56.3) 2. ESG 64.0 (59.9, 68.0), OSR 68.1% (64.2, 72.0)	ESG 90(57%), OSR 48 (51%)	5415 (61.3%) ESG, 437 (65.6%) OSR	ESG 14 (82%), OSR 8 (73%)
Black race	NR	% (95% C.I.): 1. ESG 7.5 (6.4, 8.5), OSR 3.8 (3.4, 4.5) (p=0.001) 2. ESG 2.7 (1.4, 4.1), OSR 2.7 (1.4, 4.0) (p=0.99)	NR	NR	NR
Charlson comorbidity score	NR	% (95% C.I.): 1. ESG 1.46 (1.38, 1.54), OSR 0.83 (0.80, 0.86) (p=0.00001)	NR	NR	NR

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		2. ESG 0.5 (0.41, 0.60), OSR 0.44 (0.35, 0.53) (p=0.34)			
Diabetes	NR	% (95% C.I.): 1. ESG 8.00 (6.7, 8.8), OSR 4.10 (3.7, 4.5) (p=0.001) 2. ESG 4.90 (3.0, 6.7), OSR 2.90 (1.5, 4.3) (p=0.09)	NR	684 (13.2%) ESG, 94 (8.3%) OSR (p<0.001)	NR
Myocardial infarction	NR	% (95% C.I.): 1. ESG 7.00 (6.00, 8.1), OSR 5.10 (4.6, 5.4) (p=0.001) 2. ESG 2.50 (1.2, 3.8), OSR 3.40 (1.9, 4.9) (p=0.38)	-NR	NR	NR
Congestive heart failure	NR	% (95% C.I.):	ESG 13 (9%), OSR 9 (10%) (p=1.00)	NR	NR

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		1. ESG 10.85% (9.6%-12.0%), OSR 10.60% (10.1%- 11.2%) (p=0.16) 2. ESG 3.00 (1.6, 4.5), OSR 4.70 (2.9, 6.5) (p=0.16)			
Cerebrovascular disease	NR	% (95% C.I.): 1. ESG 5.09% (4.2%-5.9%), OSR 3.20% (2.9%- 3.5%) (p=0.65) 2. ESG 2.00 (0.5, 2.6), OSR 2.00 (0.8, 3.1) (p=0.65)	NR	NR	4 (23%) ESG, 1 (9%) OSR (p=NS)
COPD	NR	% (95% C.I.): 1. ESG 22.04% (20.3%-23.6%), OSR 10.65% (10.0%-11.2%) (p=0.47)	ESG 56 (40%), OSR 36 (38%) (p=0.89)	247 (34.7%) ESG, 1483 (17.9%) (P<0.001)	10 (59%) ESG, 6 (54%) OSR (p=NS)

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		2. ESG 4.00 (2.3, 5.6), OSR 4.90 (3.0, 6.7) (p=0.47)			
Chronic renal failure	NR	% (95% C.I.): 1. ESG 4.9% (4.0%- 5.7%), OSR 0.88% (0.07%- 0.10%) (p=0.70) 2. ESG 0.01 (0, 1.4), OSR 0.50 (0, 1.1) (p=0.70)	NR	90 (12.6%) ESG, 126 (1.5%) OSR (p<0.001)	4 (23%) ESG, 1 (9%) OSR (p=NS)
History of malignancy	NR	% (95% C.I.): 1. ESG 5.71% (4.7%-6.6%), OSR 2.91% (2.6%- 3.2%) (p=0.82) 2. ESG 1.80 (0.6, 2.9), OSR 1.60 (0.5, 2.6) (p=0.82)	NR	NR	NR
History of smoking (current or past)	NR	NR	ESG 117 (84%), OSR 77 (82%) (p=0.86)	NR	NR

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BMI (kg/m2)	NR	NR	mean (SD): ESG 26.4 (4.7), OSR 26.9 (5) (p=0.44)	NR	NR
Coronary artery disease	NR	NR	ESG 69 (49%), OSR 34 (36%) (p=0.60)	NR	NR
Stroke	NR	NR	ESG 14 (10%), OSR 9 (9%) (p=1.00)	NR	NR
Peripheral arterial disease	NR	NR	ESG 22 (16%), OSR 10 (11%) (p=0.33)	NR	NR
Symptomatic aneurysm	NR	NR	ESG 30 (21%), OSR 36 (38%) (p=0.007)	NR	NR
Other concomitant aneurysm:	NR	NR	ESG 39 (28%), OSR 26 (28%) (p=1.00)	NR	NR
Renal dialysis	NR	NR	ESG 2 (1%), OSR 0 (0%) (p=0.52)	NR	NR

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Paraplegia	NR	NR	ESG 1 (1%), OSR 0 (0%) (p=1.00)	NR	NR
Cancer	NR	NR	ESG 27 (19%), OSR 12 (13%) (p=0.21)	NR	NR
Prior major vascular surgery	NR	NR	ESG 63 (45%), OSR 52 (55%) (p=0.14)	NR	NR
Cardiac comorbidities	NR	NR	NR	265 (37.2%) ESG, 3535 (42.8%) (p<0.001)	NR
Hypertension	NR	NR	NR	NR	16 (94%) ESG, 10 (91%) OSR (p=NS)
Peripheral obstructive disease	NR	NR	NR	NR	5 (29%) ESG, 2 (18%) OSR (p=NS)
Ischemic heart disease	NR	NR	NR	NR	6 (35%) ESG, 5 (45%) OSR (p=NS)

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Cardiac Surgery	NR	NR	NR	NR	(18%) ESG, 0 (0%) OSR (p=NS)
SHORT- TERM OUTCOMES					
All-cause mortality	ESG 21 (5.9%) OSR 4 (1.5%) Adjusted OR: 0.79	1. ESG 6.1% (95%CI 5.1%, 7.0%) OSR 7.1% (95%CI 6.7%, 7.6%) 2. ESG 4.2% (95% C.I. 2.5%-5.8%) OSR 4.5% (95% C.I. 2.8%-6.2%)	ESG 3 (2.1%) OSR 11 (11.7%) Adjusted OR: 0.54	ESG 25 OSR 379	ESG 1 (6%) OSR 2 (20%)
Re-intervention	ESG 21 (5.9%) OSR 4 (1.5%) Adjusted OR: 2.97 (1.09, 8.10)	NR	NR	NR	NR

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Paraplegia/ Paraperisis	NR	NR	ESG 4 (3%) OSR 13 (14%)	NR	NR
Cerebral Vascular Accident	NR	NR	ESG 5 (4%) OSR 5 (4%)	NR	ESG 2 (9%) OSR 1 (12%)
Neurological- unspecified type	NR	NR	NR	ESG 20 (2.8%) OSR 273 (3.3%) Adjusted OR: 0.38	NR
Respiratory failure	NR	NR	ESG 5 (4%) OSR 19 (20%)	NR	NR
Pneumonia	NR	NR	NR	NR	ESG 2 (12%) OSR 3 (27%)
Pulmonary- unspecified type	NR	NR	NR	ESG 17 (2.4%) OSR 462 (5.6%) Adjusted OR: 0.48	NR
Myocardial Infarction	NR	NR	ESG 0 (0%) OSR 1(1%)	NR	ESG 0 (0%) OSR 1 (9%)

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Cardiac-unspecified type	NR	NR	NR	ESG 28 (2.9%) OSR 1252 (15.2%) Adjusted OR: 0.24	NR
Peripheral vascular disease	NR	NR	ESG 20 (14%) OSR 4 (4%)	NR	NR
Renal failure	NR	NR	ESG 2 (1%) OSR 12 (13%)	NR	NR
Wound infection/dehiscence	NR	NR	ESG 5 (4%) OSR 10 (11%)	NR	NR
GI complication	NR	NR	ESG 3 (2%) OSR 6 (6%)	NR	NR
Post-implant syndrome	NR	NR	NR	NR	ESG 4(18%) OSR 0(0%)
LONG-TERM OUTCOMES					

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All-cause survival	1 year adjusted OR: 1.1 95%CI [0.70-1.73], p=0.667 5 year adjusted HR: 1.45 95%CI [1.08-1.94], p=0.013	NR	NR	NR	NR
aortic-related survival	5 year adjusted HR: 1.27 95%CI [0.81-1.98], p=0.298	NR	NR	NR	NR
survival from aortic-related re-intervention	5 year adjusted HR: 1.70 95%CI [1.11-2.60], p=0.014	NR	NR	NR	NR
* 1. Unadjusted cohort, 2. Propensity matched cohort					

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Appendix 3: Risk of Bias assessment

Risk of Bias Results using the ROBINS-I risk of bias tool

Domain – source of bias	Goodney et al 2011 *(2 analyses*)*	Gore TAG	Hughes et al 2014	Piffaretti et al 2007	Von Allmen et al 2014
Confounding	1. Severe 2. Moderate	Severe	Severe	Severe	Severe
Selection of participants into the study	1. Low 2. Low	Low	Low	Low	Low
Classification of interventions	1. Low 2. Low	Low	Low	Low	Low
Deviations from intended interventions	1. Low 2. Low	Low	Low	Low	Low
Missing data	1. Low 2. Low	Low	Low	Low	Low
Measurement of outcomes	1. Low 2. Low	Low	Low	Low	Low
Bias in selection of the reported result	1. Moderate 2. Moderate	Moderate	Moderate	Moderate	Moderate
Overall	1. Severe 2. Moderate	Severe	Severe	Severe	Severe
* 1. Unadjusted cohort, 2. Propensity matched cohort					

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8

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Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency ($e.g., I^2$) for each meta-analysis.	9

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1, appendix 2
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	14, appendix 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	14-19
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	14-19
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	14-19
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA

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DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	19-21
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	21
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	22
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.

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A Systematic Review of Endovascular Stent Grafting Versus Open Surgical Repair for the Elective Treatment of Arch/Descending Thoracic Aortic Aneurysms

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A Systematic Review of Endovascular Stent Grafting Versus Open Surgical Repair for the Elective Treatment of Arch/Descending Thoracic Aortic Aneurysms

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ABSTRACT

Objective: to review comparisons of the effectiveness of endovascular stent grafting (ESG) against open surgical repair (OSR) for treatment of chronic arch or descending thoracic aortic aneurysms (TAA).

Design Systematic review and meta-analysis

Data sources MEDLINE, EMBASE, CENTRAL, WHO International Clinical Trials Routine data collection, current controlled trials, clinical trials, and the NIHR portfolio were searched from January 1994 through March 2020.

Eligibility criteria for selective studies All identified studies that compared ESG and OSR, including RCTs, quasi-randomized and non-randomized controlled trials, comparative cohort studies and case-control studies matched on main outcomes were sought. Participants had to receive elective treatments for arch/descending (TAA). Studies were excluded where other thoracic aortic conditions (e.g. rupture or dissection) were reported, unless results for patients receiving elective treatment for arch/descending TAA reported separately.

Data extraction and synthesis Data was extracted by one reviewer and checked by another. Risk of Bias was assessed using the ROBINS-I tool. Meta-analysis was conducted using random effects. Where meta-analysis not appropriate, results were reported narratively.

Results Five comparative cohort studies met inclusion criteria, reporting 3955 ESG and 21,197 OSR patients. Meta-analysis of unadjusted short-term (30 day) all-cause mortality favoured ESG (OR: 0.75; 95%CI (0.55, 1.03)). Heterogeneity identified between larger and smaller studies. Sensitivity analysis of four studies including only descending TAA showed no statistical significance (OR 0.73, 95 %CI. (0.45, 1.18)), moderate heterogeneity. Meta-analysis of adjusted short-term all-cause mortality favoured ESG (OR 0.71, 95%CI (0.51, 0.98)), no heterogeneity. Longer-term (beyond 30 days) survival from all-cause mortality favoured OSR in larger studies and ESG in smaller studies. Freedom from re-

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intervention in the longer-term favoured OSR. Studies reporting short-term non-fatal complications suggest fewer events following ESG.

Conclusions There is limited and increasingly dated evidence on the comparison of ESG and OSR for treatment of arch/descending TAA

PROSPERO registration number CRD42017054565

KEY WORDS

Systematic Review, Meta-analysis, Thoracic Aortic Aneurysm, Open Surgical Repair, Endovascular Stent Graft

STRENGTHS AND LIMITATIONS OF THIS STUDY

Strength:

- This is the first systematic review to investigate two-armed studies looking only at elective ESG versus OSR for treatment of chronic arch or descending thoracic aortic aneurysms. Other reviews include dissections, ruptures and treatment for other conditions.
- This systematic review inclusion criteria were broad and sought to include evidence from both randomized and non-randomized study designs.

Limitation:

- Statistical analysis of non-fatal complications was not possible due to the limited and inconsistent reporting of such complications.

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INTRODUCTION

Rationale

A thoracic aortic aneurysm (TAA) is an abnormal expansion of the aorta at any part of its course from the heart to the diaphragm. Aneurysms manifest with acute symptoms or they may be chronic. If untreated, TAA can continue to expand and the annual risk of aneurysm rupture increases from 2% in aneurysms smaller than 5cm in diameter to 7% in aneurysms over 6cm in diameter[1]. The majority of aneurysms (and therefore studies) centre on the ascending or abdominal segments of the aorta, arch and descending aneurysms are relatively under-studied. Therefore, this review focused on chronic aneurysms affecting the arch or descending thoracic aorta.

There are currently two main methods of repair for arch/descending TAA: Open Surgical Repair (OSR) and Endovascular Stent Grafting (ESG). OSR involves replacing the aneurysmal segment of aorta with a surgical graft during open surgery using heart-lung bypass. As the arch and descending aorta supply blood to the brain and spinal cord, OSR is high risk, with reported mortality and paraplegia rates of approximately 5% and 10% [2] respectively. In ESG a stent is placed across the aneurysm, providing a new conduit through which the blood flows. ESG is most often selected for patients who are considered at high risk of death or permanent disability from OSR. However, some patients may not have a suitable anatomy/morphology for ESG procedures [2]. Concerns have also been raised about long-term durability and safety of the implanted devices [3 4] due to complications such as stent migration or endoleaks, with the incidence of late endoleaks (leaks around or through the stent graft that occur more than one year after insertion) reported as being between 2.5% and 27.8% [5]. The need for re-intervention due to technical failures of the stent following ESG is also high (23% at 5 years) [6 7]. With each re-intervention there is an added risk of complication either due to the increased complexity of the procedure or deteriorating health of the patient [2].

One previous systematic review comparing only elective OSR and ESG for descending and arch TAAs has been reported [8]. This systematic review [8] was restricted to data from randomized controlled

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trials in which patients were randomly assigned to ESG and OSR, but no eligible trials were identified when searching relevant data based up to Jan 2016. There are other systematic reviews [9 10] that have included non-randomized studies; however, these reviews also included emergency cases and other thoracic aortic pathologies (e.g. dissection or rupture). As such, they may not represent a true comparison of elective ESG and OSR procedures for descending thoracic aortic aneurysms. Given the lack of RCT evidence, this systematic review includes non-randomized comparative studies of the relative effectiveness of OSR or ESG.

The decision to operate requires a favourable balance between the risks and benefits of that intervention compared to those attached to the expected outcomes of the disease. This balance is in turn influenced by a number of factors. Aneurysm size correlates to the risk of death (if left untreated) and so patients are usually referred once the aneurysm crosses a chosen threshold. At that point, patient fitness (*and life expectancy*), the reality or possibility of an intrinsic weakness affecting the whole aorta, the technical feasibility of the proposed intervention and patient wishes, come into play. This systematic review takes a step back from these issues and examines only the outcomes for patients in whom a decision to intervene has been made. This review is an essential starting point for the analysis of effective treatments for thoracic aortic aneurysms.

Objectives

The aim of this systematic review was to assess the effectiveness of endovascular stent grafting (ESG) compared to open surgical repair (OSR) as a treatment for chronic arch or descending TAA.

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METHODS

Protocol and registration

The review protocol was registered in the PROSPERO database (CRD42017054565)[11].

PUBLIC PATIENT INVOLVEMENT

There was no patient and public involvement in the whole process of conducting this review.

Eligibility criteria

Study designs

All identified studies that directly compared ESG and OSR, including RCTs, quasi-randomized and non-randomized controlled trials, comparative cohort studies and case-control studies matched on main outcomes were sought [12].

Non-randomized studies that controlled for selection bias were included – irrespective of study size. Acceptable methods for addressing selection bias included matching patients at baseline for important covariates, matching patients using propensity scores or adjustment for important covariates during analysis. Non-randomized studies that did not use these methods but included larger sample sizes (more than 250 patients) were also included as they have the potential to reduce sampling variation relative to smaller unmatched studies[13].

Non-English language papers were noted and, where possible, data were extracted from the abstract. No translation of full text articles was possible.

Study Participants

Study participants had to receive elective treatments for arch and/ or descending thoracic aortic aneurysms. Where emergency interventions or other thoracic aortic conditions (such as intramural haematoma, rupture, and dissection) were reported, studies were excluded, unless results for patients receiving elective treatment for arch/descending TAA were reported separately. Studies

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that reported secondary (re-intervention) procedures for arch/descending TAAs were included only if the original, primary procedure for each participant could be identified.

Study interventions

Open Surgical Repair was defined as replacement of the aneurysmal aorta with a prosthetic conduit via a sternotomy or thoracotomy with circulatory support. Endovascular stent grafting was defined as transluminal introduction of a stent-graft under X-ray guidance.

Outcome measures

This review assessed short-term and long-term clinical effectiveness. Short-term was defined as either within 30-days of the intervention and/or time until discharge following the intervention. Since most studies did not report long-term outcomes conditional on surviving the short-term period, long-term analyses covered the time from intervention until the end of follow-up. Primary outcomes were all-cause mortality and aneurysm-related mortality. Secondary outcomes were re-interventions and non-fatal complications. The latter were sub-classified as vascular, neurological, cardiac, respiratory and 'other' complications.

Information sources

MEDLINE, EMBASE, CENTRAL, WHO International Clinical Trials Routine data collection, Current Controlled Trials, Clinical Trials, and the UK National Institute for Health Research portfolio were searched. The search strategy (Appendix 1) was adapted as appropriate for each database (for example to allow for variations in controlled vocabulary terms and syntax [14]). Electronic searches identified publications between 1994, date of first reported case of ESG for descending and arch TAA[15], and March 2020.

Websites of regulatory bodies and Health Technology Assessment agencies were screened for unpublished relevant reports. A citation search of all included studies was conducted via Web of

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Science. Studies from systematic reviews that satisfied the inclusion criteria of this review were screened for eligibility.

Study selection

All titles and abstracts were screened by one reviewer (AM). A second investigator (JG) reviewed 10% of all titles and abstracts classed as eligible or not eligible and all titles and abstracts classified as unclear by the first reviewer. The full text of selected studies was assessed by one investigator (AM) and classified as relevant, not relevant or unclear according to the inclusion criteria. A second investigator (JG) reviewed 10% of all papers, as well as all studies deemed unclear. Disagreements at either stage were resolved by discussion and where necessary a third reviewer (PM) arbitrated. Where required, a clinical expert (PS) was asked to screen papers.

Data collection process

A data extraction form was developed based on The Cochrane Collaboration data collection form [16]. Data were extracted by one reviewer (AM) and checked by a second reviewer (JG) for accuracy. Disagreements were resolved by discussion or arbitration by a third reviewer (PM).

Data items

Reported short-term and longer-term all-cause mortality, as well as short-term and longer-term aneurysm-related mortality were collected. All reported complications were recorded and grouped according to expert clinical guidance as described above as neurological, cardiac, respiratory or 'other' complications, with endoleaks following ESG recorded separately.

Risk of bias in individual studies

Risk of bias tools used were specific to the study design. As described later only non-randomized study designs were identified. Therefore, the ROBINS-I tool [17] was used. The ROBINS-I tool classifies the risk of bias in seven domains (*confounding; selection of participants; classification of*

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interventions; deviations from intended interventions; missing data; measurement of outcomes; selection of the reported results) and overall as either critical, severe, moderate, low, or unclear.

Synthesis of results

Point estimates and precision of the incidence of short and long-term events and unadjusted and adjusted comparisons (odds ratios (OR) hazard ratios (HR)) were recorded. Meta-analysis of short-term event-comparisons was based on the OR with 95% confidence interval (CI) and conducted using RevMan [18]. An overall treatment effect was calculated for the log (OR) of each outcome using a (normal) random effects model. Where comparisons could not be made, data were tabulated and reported narratively.

Heterogeneity between studies was assessed according to Cochrane guidelines, on clinical criteria, by visual inspection of plots of the data, from the chi-square test for heterogeneity, and the I-squared statistic. An I^2 statistic of 25-50% indicated low heterogeneity; 50-75% indicated moderate heterogeneity, and over 75% indicating significant heterogeneity.

RESULTS

Study selection

The search identified 5163 unique studies. Of these, 5078 studies were not eligible. Ineligibility was primarily because of irrelevant anatomy (ascending, abdominal, or peripheral aneurysms), irrelevant pathology (acute dissection or other acute aortic syndromes), or because the study pooled data from a mixture of anatomical and pathological situations such that data on chronic arch and DTA aneurysms could not be extracted with confidence. Thus 85 papers underwent full text screening. 75 papers were rejected at this stage leaving 10 papers to be included in the full review. Six papers reported data from the Gore TAG Trial [19-24], leaving five unique comparative studies that fulfilled the inclusion criteria. Further details of the selection process are shown in the PRISMA diagram (Figure 1).

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One study, Goodney et al 2011[25], reported both an unadjusted analysis for the unadjusted cohort and an analysis of a subgroup of propensity score matched patients. Data from the unadjusted cohort were used in all meta-analyses except the adjusted mortality meta-analysis, which used the propensity matched data. Brief summaries of the identified studies are given in Table 1, with detail provided in Appendix 2. Overall, the five studies included data on 3955 ESG and 21,197 OSR patients and all studies were conducted in Europe or the USA. Two studies had non-overlapping or only partially overlapping recruitment periods between the ESG and OSR intervention groups (Gore Tag[19-24] and Piffaretti et al 2007[26]). There were clear differences in patient characteristics between the two treatment groups. ESG patients were consistently older, with average age differences ranging from 2 to 9 years in all but one analysis (Goodney et al 2011[25] the propensity matched cohort). Similarly, in most studies the proportion of men was greater in the ESGgroup.

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Table 1: Study characteristics

Study	**Von Allmen et al 2014**	**Goodney et al 2011 (2 analyses*)**	**Gore TAG**	**Hughes et al 2014**	**Piffaretti et al 2007**
METHODS					
Study Design	Comparative cohort	Comparative cohort	Comparative cohort	Comparative cohort	Comparative cohort
Method of matching	None	1. None 2. propensity matched	ESG patients judged eligible for OSR	None	OSR patients judged eligible for ESG
PARTICIPANTS					
Country	UK	USA	USA	USA	Italy
Setting	Multi-centre – Hospital episode statistics	Multi-centre – Medicare data	Multi-centre	Multi-centre	Single-centre
Time frame	2006 to 2011	1. 1998-2007 2. 2003-2007	ESG 1999-2001, OSR pre 1999 (n=44) & 1999-2001 (n=50)	1998-2007	ESG 2000-2007, OSR 1996-2000
Recruitment	Total 618 ESG 354 OSR 264 OSR	1. Total 13,998 (ESG 2433, OSR 11,565)	Total 234 ESG 140 OSR 94	Total 8967 ESG 712 OSR 8255	Total 28 ESG 17 OSR 11

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		2. Total 1100 (ESG 550, OSR 550)			
Age summary	median (IQR) ESG 73(66,78), OSR 71 (63-76)	mean (95% C.I.) 1. ESG 75.9 (75.6, 76.1), OSR 73.8 (73.7, 73.9) 2. ESG 71.1 (70.8, 71.4), OSR 70.7 (70.7, 71.1)	mean(SD) ESG 75.9 (10.4), OSR 68.2 (10.2)	median (IQR) ESG 72 (63-78), OSR 63 (52, 72)	mean (SD) ESG 66 (10), OSR 61 (13)
Proportion male	ESG 232 (65.5%), OSR 137 (51.9%)	% (95% C.I.): 1. ESG 58.7 (56.7, 60.7), OSR 55.4 (54.8, 56.3) 2. ESG 64.0 (59.9, 68.0), OSR 68.1% (64.2, 72.0)	ESG 90(57%), OSR 48 (51%)	5415 (61.3%) ESG, 437 (65.6%) OSR	ESG 14 (82%), OSR 8 (73%)
OUTCOMES					
Short-term	Death from any cause	Death from any cause	Death from any cause	Death from any cause	Death from any cause

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	Re-intervention All-cause mortality (unadjusted		Complications	Complications	Complications
Long-term	Time to death from any cause Time to aortic-related death Time to aortic-related re-intervention	Time to death from any cause	Time to death from any cause Time to aneurysm-related death aneurysm-related mortality Time to re-intervention Time to major adverse event	Not reported	Time to death from any cause
* 1. Unmatched cohort, 2. Propensity matched cohort					

Risk of bias within studies

The propensity-matched analysis of Goodney et al (2011) [25] was judged to be of overall moderate risk of bias and four studies were at severe risk of bias due to the potential for confounding (Appendix 3).

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Two large cohorts derived from UK and USA administrative databases (Von Allmen et al 2014[27] and Hughes et al 2014[28]) did not attempt to address selection bias at the design stage, but did adjust for some confounders in the analysis. A third study (Goodney et al 2011[25]), using USA administrative data, included both unadjusted population and analysis of a subset of the data defined by propensity score matching. Propensity score methods rely on all important confounders being available and included[29]. As this study did not include all confounders considered important, such as aneurysm size, this study was judged to have moderate risk of bias due to confounding.

One study (Gore TAG)[19-24] included only ESG patients who would also have been eligible for OSR and the another study (Piffaretti et al 2007)[26] included only OSR patients who would be suitable for ESG, both according to clinician judgement. Since suitability was decided retrospectively, such matching may not be robust.

A further potential bias with each study was selective reporting. Although the measurements and analyses of the outcomes were clearly defined within each study, no evidence of a pre-study analysis plan was identified.

Synthesis of results

Short-term death from any cause (30-day or in hospital)

Unadjusted Mortality:

All five studies[19-28] reported unadjusted 30 day or in-hospital mortality (Figure 2). Overall, the odds of short-term mortality were lower in the ESG group than the OSR group (OR: 0.75; 95 %CI. (0.55-1.03)). Moderate heterogeneity was present in the overall analysis (I^2= 40%), mainly due to differences between the two smaller studies[19-24 26] and the three larger studies [25 27 28], with a large I^2 for subgroup differences between the larger studies subgroup and the smaller studies subgroup of 84.5%.

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For the three larger studies[25 27 28] (3798 ESG and 21,092 OSR patients), the treatment effect for death was smaller (OR: 0.84 95%CI (0.72, 0.98)) than that for the two smaller studies[19-24 26] (157 ESG and 105 OSR patients, combined OR: 0.18 95%CI (0.06, 0.59)). There was no evidence of heterogeneity in results between these subgroups (I^2=0).

Sensitivity analysis

Hughes et al 2014[28] was the only study to include aortic arch aneurysms. Arch aneurysms tend to be more complex procedures and are mostly conducted using OSR. A sensitivity analysis excluding this study was therefore conducted (Figure 3).

While the point effect estimates all favoured ESG, the overall pooled odds ratio (OR: 0.73; 95 %CI. (0.45-1.18)) did not reach statistical significance (p=0.2). Moderate heterogeneity was present in the overall analysis (I^2= 42%), mainly due to differences between the two smaller studies and the two larger studies, with a large I^2 for subgroup differences between the larger studies subgroup and the smaller studies subgroup of 65.1%.

For the two larger studies[25 27] (3086 ESG and 12837 OSR patients), the treatment effect for death was smaller (OR: 0.85 95%CI (0.72, 1.00)) than that for the two smaller studies[19-24 26] (157 ESG and 105 OSR patients, combined OR: 0.20 95%CI (0.04, 1.06)). There was no evidence of heterogeneity in results within the large subgroup (I^2= 0) and evidence of low heterogeneity withing the small studies subgroup (I^2=26%).

Adjusted Mortality:

The three larger studies [25 27 28] included adjusted analysis of short-term mortality (Figure 4). Goodney et al 2011[25] used stratified propensity matching. Hughes et al 2014 [28] adjusted for preoperative comorbidities, independent predictors of mortality and post-operative complications and the Von Allmen 2014[27] study adjusted for age and sex. The OR for ESG, relative to OSR, for these three

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studies combined was 0.71, 95%CI (0.51, 0.98). No evidence of heterogeneity was identified in this meta-analysis (I^2=0%)

All-cause mortality (following hospital discharge or after 30 days)

Meta-analysis of the longer-term all-cause mortality was not possible as follow-up varied between the four studies that reported this outcome (Gore Tag[19-24], Goodney et al 2011[25], Piffaretti et al 2007[26] and Von Allmen 2014[27]) . The results in the larger studies (Goodney et al 2011[25], Von Allmen 2014[27]) differed from those in the smaller studies (Gore Tag[19-24] and Piffaretti et al 2007[26]). We note that the analyses included the short-term deaths as well as those occurring beyond hospital discharge or after 30 days.

Von Allmen et al 2014[27] reported time to death from any cause up to five years, throughout which OSR had higher overall survival. Adjusting for age and sex, ESG had higher odds of death at one year (OR: 1.10 95%CI (0.70, 1.73); p=0.667). Additionally, the hazard ratio for ESG relative to OSR up to five years, adjusting for age and sex, was HR=1.45 (95%CI [1.08, 1.94], p=0.013).

Goodney et al 2011[25] also reported time to death from any cause up to five years. Adjusting for age, sex, race, era of procedure and Charlson comorbidity score, five-year survival for OSR in the unmatched cohort was higher than for ESG (89% versus 79%, log-rank p<0.0001). In the propensity-matched cohort, five-year life-table analysis reported higher survival in the OSR group (81% 95%CI (77%-85%)) than for the ESG group (73% 95%CI (68%, 76%)) (log-rank p=0.007).

In the Gore TAG study[19-24], Time to death from any cause over five years did not differ between the ESG and OSR groups, with survival probability at five years in both groups of 63% (log rank P value= 0.625). Piffaretti et al 2007[26] reported survival overall in the ESG group up to five years and the OSR group up to 13 years. For the five years in which survival is reported in both groups, ESG had a higher

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survival compared to OSR although this was based on only 4 deaths, and the follow-up periods differed substantially (log rank p>0.05).

Aneurysm/Aortic-related mortality

Meta-analysis of Aneurysm/aortic-related mortality could not be conducted. No studies reported aneurysm-related mortality during hospitalization or up to 30 days separately to the long-term. Von Allmen et al 2014[27] defined aortic-related mortality as any aortic-related death occurring within five years of the index operation according to ICD-10 codes referring to aorta, or cause of deaths closely related to complications of surgery (e.g. paraplegia). Time to aortic-related death did not differ between OSR group than the ESG group during the five-year follow-up (log-rank p=0.270). This study[27] also reported adjusted (for age and sex) five year HR of 1.27 (0.81 to 1.98; P =0.298) for ESG patients relative to OSR patients.

Gore TAG[19-24] defined aneurysm related mortality as deaths that occurred in hospital or within 30 days of initial procedure or any reintervention or a death due to the aneurysm or the treatment device and reported up to five years, with longer time to aneurysm-related death in the ESG group over this period (log-rank p= 0.012).

Short-term re-interventions

One study reported aortic-related re-interventions in 21 (5.9%) ESG patients and 4 (1.5%) OSR patients, an adjusted OR of 2.97 (95%CI [1.09, 8.10], p=0.033) for ESG versus OSR (Von Allmen et al 2014[27]).

Long-term re-interventions

Two studies reported this outcome (Von Allmen et al 2014[27] and Gore TAG [19-24]). Von Allmen et al 2014[27] reported time to aortic-related re-interventions up to five years. Time to re-intervention was

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longer in the OSR group compared to the ESG group (log-rank p=0.007), with adjusted (for age and sex) hazard ratio at five years of 1·70 (95%CI [1·11 to 2·60], p =0·014). In Gore TAG[19-24], time to re-intervention over five years was longer in the ESG group (log-rank P=0.011). We note that these analyses included short-term re-interventions.

Complications

Table 2 summarises the short-term complications reported in all five studies[19 25-28]. Most studies reported a small number of neurological, pulmonary, cardiac and other complications. However, one larger study (Hughes et al 2014)[28] reported lower odds for ESG patients for neurological complications, OR of 0.48 (95%CI (0.26, 0.86), p=0.015), pulmonary complications, OR: 0.38 (95%CI (0.21, 0.67), p=0.001) and cardiac complications, OR: 0.24 (95%CI (0.15, 0.37), p<0.001). The Gore TAG study[19-24] reported endoleak rates for ESG patients of 8.5%.

Table 2: Short-term complications

Study	**Gore TAG**		**Piffaretti et al 2007**		**Hughes et al 2011**	
Arm (participants)	**ESG (n=140)**	**OSR (n=94)**	**ESG (n=17)**	**OSR (n=11)**	**ESG (n=712)**	**OSR (n=8255)**
NEUROLOGICAL						
Paraplegia	NR	NR	NR	NR	NR	NR
Paraplegia/ Paraperisis	4 (3%)	13 (14%)	NR	NR	NR	NR
Cerebral Vascular Accident	5 (4%)	5 (4%)	2 (9%)	1 (12%)	NR	NR
Neurological-unspecified type	NR	NR	NR	NR	20 (2.8%)	273 (3.3%)
RESPIRATORY						

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Respiratory failure	5 (4%)	19 (20%)	NR	NR	NR	NR
Pneumonia	NR	NR	2 (12%)	3 (27%)	NR	NR
Pulmonary-unspecified type	NR	NR	NR	NR	17 (2.4%)	462 (5.6%)
CARDIAC						
Myocardial Infarction	0 (0%)	1(1%)	0 (0%)	1 (9%)	NR	NR
Cardiac-unspecified type	NR	NR	NR	NR	28 (2.9%)	1252 (15.2%)
ENDOLEAKS						
Endoleaks	12 (8.5%)	NR	NR	NR	NR	NR
OTHER						
Peripheral vascular disease	20 (14%)	4 (4%)	NR	NR	NR	NR
Renal failure	2 (1%)	12 (13%)	NR	NR	NR	NR
Wound infection/dehiscence	5 (4%)	10 (11%)	NR	NR	NR	NR
GI complication	3 (2%)	6 (6%)	NR	NR	NR	NR
Post-implant syndrome	NR	NR	4(18%)	0(0%)	NR	NR
Note: Goodney et al 2011 and Von Allmen et al 2014 did not report short-term complications						

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DISCUSSION

This is the first systematic review that reports results from non-randomized comparative studies regarding the effectiveness of elective ESG and OSR for arch and or descending TAA. Previous reviews have either included single arm studies only[30] or were comparative reviews that included non-elective procedures[9 31], or mixed pathologies such as dissections[10 32]. The results of these

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reviews may not be applicable to patients undergoing elective procedures for descending or arch TAA pathology.

In this review only five comparative cohort studies were identified. The lack of RCTs comparing ESG and OSR for elective arch/descending thoracic aortic aneurysms may reflect the lack of equipoise from practitioners and patients. Well-designed non-randomized trials can demonstrate a high degree of concordance with the results of well-designed RCTs, provided that all confounders are adequately accommodated in the design and/or analysis[33]. However, the studies identified in this review had moderate to serious risk of bias, mainly due to potential for confounding.

Although, complications in the short-term also appear to favour ESG, evidence regarding short-term complications is either lacking or inconsistently reported. All five studies[19 25-28] suggested that the less invasive nature of ESG confers lower short-term mortality and fewer short-term non-fatal complications than OSR. The two small unadjusted studies[19-24 26] had a much larger effect than larger studies[25 27 28]. These larger studies used more robust methods for addressing confounding; this may indicate the presence of publication bias where only those smaller studies with larger effect sizes have been published.

Longer-term results are difficult to interpret since most data were not presented in a way to separate early risk from longer term survival. The two smaller studies[19-24 26] reported continuing survival benefit for ESG patients in the long-term but are limited by small numbers, different recruitment periods, and failure to exclude early risk. The survival advantage associated with ESG in the Gore TAG[19-24] study disappears by year five, which suggests that the hazards are lower for OSR patients that survive the initial procedure. Conversely, the two larger studies (Von Allmen et al 2014[27] and Goodney et al 2011[25]) report that the short-term survival advantage of ESG were reversed by twelve months, and that the OSR group had an increasing survival advantage thereafter. Similarly, time to re-intervention favoured ESG in the smaller Gore TAG[19-24] study, whilst time to aortic-related re-interventions favoured OSR in the larger study Von Allmen et al 2014[27].

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The evidence regarding mortality is inconsistent across the studies with more recent, larger studies[25 27 28] reporting a more modest survival benefit for ESG in the short-term and a survival benefit for OSR in the long-term. The two smaller studies [19-24 26]suggest a large reduction in the risk of mortality and short term complications in the short run and a survival benefit in the long-run for ESG.

Inconsistency of results across studies may be due to heterogeneity of the study populations, both within and across studies. Differences in recruitment periods may reflect changes in technology and practice in both groups. The larger studies [25 27 28] were based on patients recruited up to twenty years later than the smaller studies. Outcomes may have improved differentially with OSR patients receiving proportionally better outcomes due to general improvements in surgical practice and patient selection over time[34]. A potential reason for the reported long-term survival benefit of OSR in more recent, larger studies may be differences in patient risk in each intervention group. Open surgical repair is advocated for younger, lower-risk patients, and EVAR is preferred for older, higher-risk patients[35]. This selection bias may not have been adequately accommodated in adjusted analyses.

The three large studies[25 27 28] were based on administrative health records with case ascertainment that depends on accurate coding, which can vary across and within providers[36]. Furthermore, one of the small studies, Piffaretti et al 2007[26], recruited patients from one centre which may have compromised external validity as they may for example, depend on the expertise within a centre and the nature of the patients that centre treats[37 38]. Additionally, the Gore TAG study[19-24] utilised a more restricted inclusion criteria for ESG patients compared to those who received OSR, potentially introducing heterogeneity between the two groups.

The headline conclusion of this review is that whilst there is a degree of variability, OSR is better for long-term survival and avoidance of re-intervention, and that ESG is better for early outcomes. This conclusion is based on non-randomized comparisons in which the influence of important confounders for choice of intervention (ESG or OSR) has not been adjusted for. Deeper exploration is

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required, but randomized data may be difficult to acquire as there are well-established guidelines about which patients are favoured for OSR and which for ESG. Thus, to compare ESG and OSR it is first necessary to establish an understanding of the reasons driving clinical management, for example by a structured consensus analysis such as a Delphi study. Thereafter, unless there is clinical equipoise about which patients are best served by OSR or ESG, the best quality evidence to establish the relative merits of ESG and OSR for arch/ DTA aneurysms is likely to be a systematic, prospective, observational, comparative study, using individual patient data.

Limitations

The findings in this review are limited by the number and potential for confounding of the five studies[19-28] identified, so that definitive conclusions are not possible. The adjustment for confounders is likely to be only partial and some bias in the estimates of differences in mortality and complications may persist, despite attempts to reduce bias in the larger studies identified. Statistical analysis of non-fatal complications was not possible due to limited and inconsistent reporting.

CONCLUSIONS

This systematic review is the first to consider evidence from non-randomized studies that directly compare ESG and OSR for the treatment of elective arch/descending thoracic aortic aneurysms. Other reviews conducted in this area have been complicated by the inclusion of treatments for emergency cases or other thoracic aortic pathologies (e.g. thoracic aortic ulcer, or dissection).

There is limited and increasingly dated evidence addressing the comparison of ESG and OSR in the management of elective arch and or descending thoracic aortic aneurysms. Further, high quality, evidence is required. Although large RCTs may not be possible in this relatively rare condition, future studies should use strong methodological study design in order to control for potential confounders and reduce potential biases. Additionally, given the current conflicting evidence, it is important that comparisons of OSR and ESG are conducted in both the short-term and the long-term.

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CONFLICT OF INTEREST

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AUTHOR CONTRIBUTIONS

Author contributions according to the CrediT (contributor roles taxonomy). Conceptualization- AM, JG, PS, LS, LV, AC, PM, SL; Data curation- AM ;Formal Analysis- AM, JG, PM; Funding acquisition- JG, AC, LV, CF, PS, LS, SL; Investigation - AM, JG, PM, LV, PS; Methodology - AM, JG, PM, LV, AC, CF, PS, LS, SL; Supervision - JG, PM, LV; Validation – AM, JG, PM, PS, LS, PC, SL; Writing – original draft - AM, JG, PM; Writing – review & editing - AM, JG, PM, AC, LV, CF, PS, LS, PC, SL.

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DATA SHARING AGREEMENT

Data were extracted from the studies included in this Systematic Review and referenced below.

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Figure 1: PRISMA Diagram

Figure 2: Forest plot of unadjusted short-term all-cause mortality

Figure 3: Forest plot of unadjusted short-term all-cause mortality. Sensitivity Analysis.

Figure 4: Forest plot of adjusted short-term all-cause mortality

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Figure 1: PRISMA Diagram

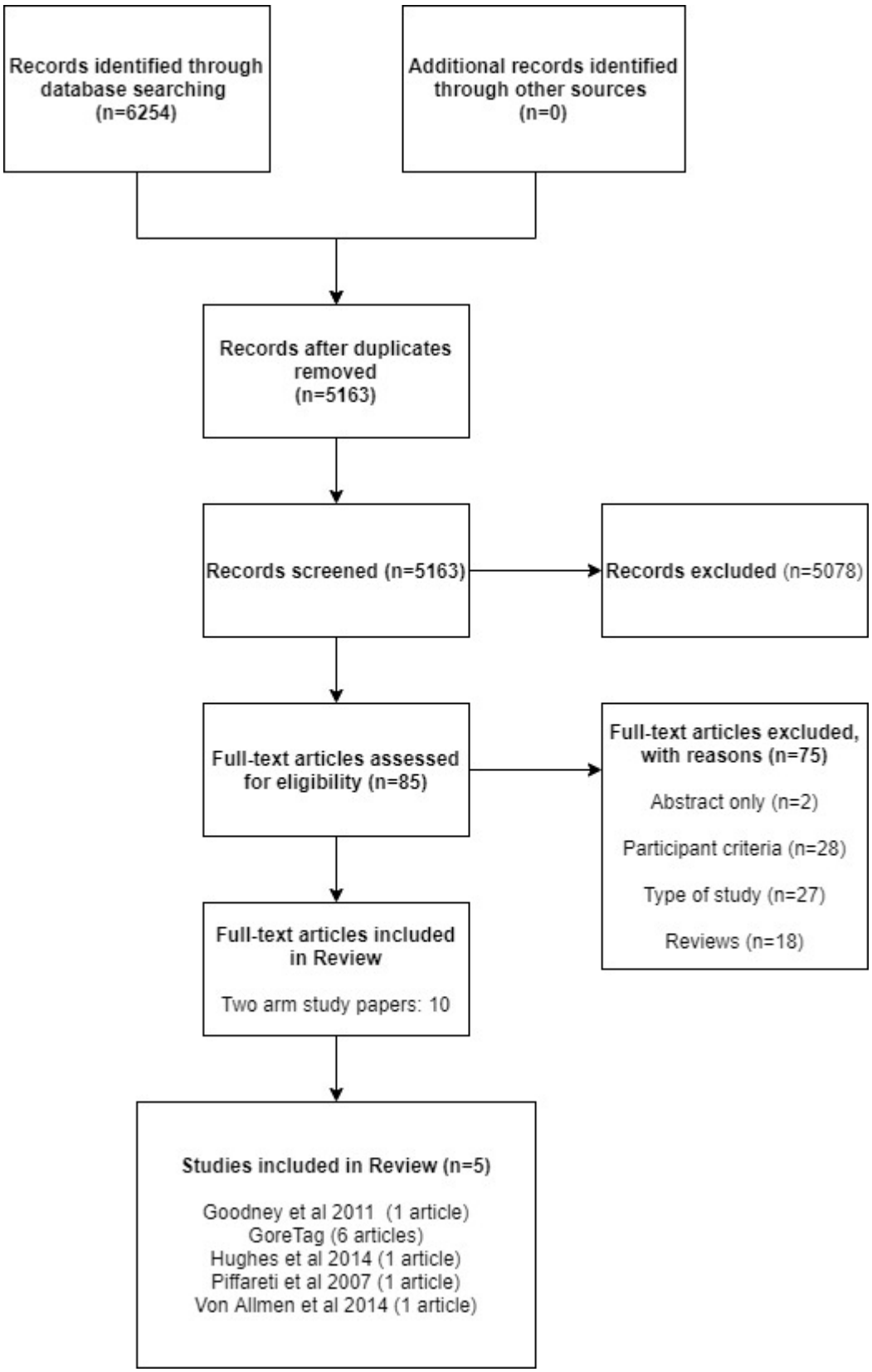
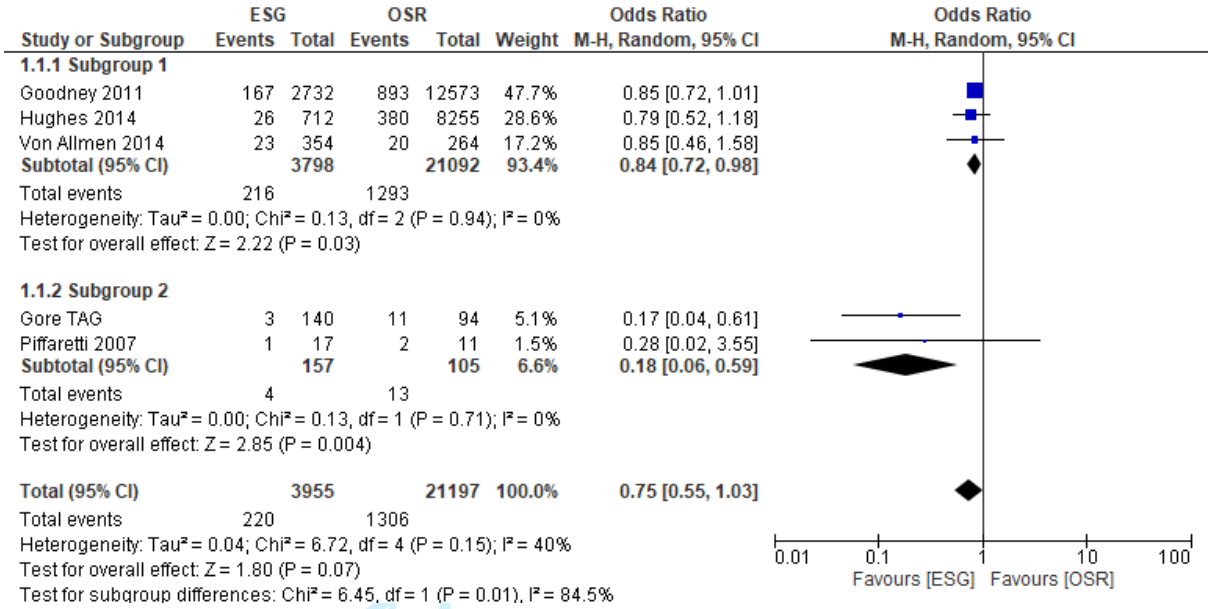
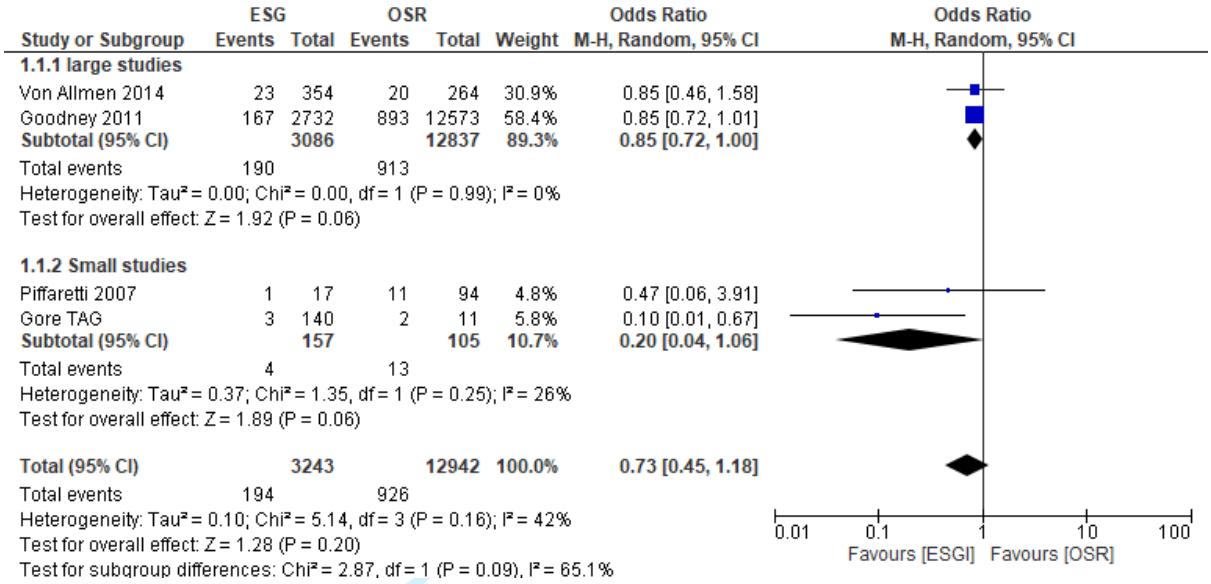


Figure 2: Forest plot of unadjusted short-term all-cause mortality



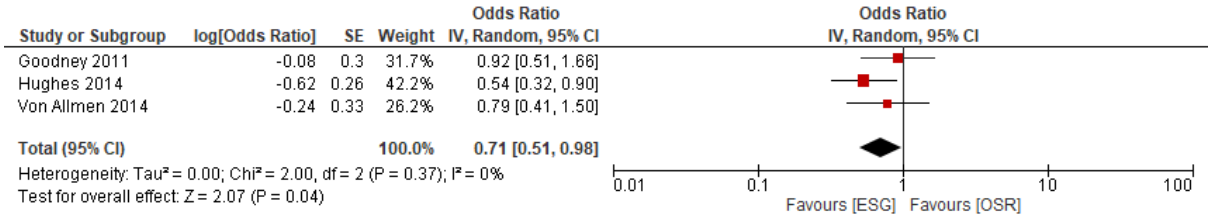
Peer review only

Figure 3: Forest plot of unadjusted short-term all-cause mortality. Sensitivity Analysis.



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Figure 4: Forest plot of adjusted short-term all-cause mortality



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Appendix 1: Search Strategy

Medline/Central/HTA

1. (TEVAR or EVAR).ti,ab.
2. (endovascular adj1 (stent* or repair* or treat* or surg* or (aneurysm? adj1 repair*))).ti,ab.
3. (open adj2 (surg* or repair*)).ti,ab.
4. endoprosthe*.mp.
5. *Aortic Aneurysm, Thoracic/su [Surgery]
6. *Aortic Thoracic/su [Surgery]
7. Thoracic Surgical Procedures/
8. or/1-7
9. exp Aortic Aneurysm, Thoracic/
10. (thorac* adj2 aneurysm*).ti,ab.
11. (descending adj1 aort* adj1 aneurysm*).ti,ab
12. TAA*.ti,ab.
13. (arch adj3 aneurysm*).ti.ab.
14. or/9-13
15. morbid*.mp.
16. mortalit*.mp.
17. outcome?.mp.
18. complication*.mp.
19. (rupture? or endoleak?).mp.
20. device migration.mp.
21. (ae or co or de).fs

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22. (safe or safety or side effect* or undesirable effect* or treatment emergent or tolerability or toxicity or adrs or (adverse adj2 (effect or effects or reaction or reactions or event or events to outcome or outcomes))).ti,ab.
23. ((technical or surg*) adj2 (advance* or improve*)).mp.
24. (quality adj2 life).mp.
25. Mortality/
26. Mortbidity/
27. Postoperative complications/
28. Intraoperative complications/
29. Aortic Aneurysm, Thoracic/co
30. Conversion to Open Surgery/
31. “Quality of Life”
32. or/15-31
33. 8 and 14 and 32
34. (“clinical trial” or “clinical trial, phase I” or “clinical trial, phase ii” or clinical trial phase iii or clinical trial, phase iv or controlled clinical trial or “multicentre study” or “randomized controlled trial”).pt. or double-blind method/ or clinical trials, phase iii as topic/ or clinical trials, phase iv as topic/ or controlled clinical trials as topic/ or randomized controlled trials as topic/ or early termination of clincal trials as topic/ or multicenter studies as topic/ or ((randomi?ed adj7 trial*) or (controlled adj3 trial*) or (clinical adj2 trial*) or ((single or doubl* or tripl* or treb*) and (blind* or mask*))).ti,ab,kw. or (“4 arm” or “four arm”).ti,ab,kw.
35. cohort studies/ or longitudinal studies/ or follow-up studies/ or prospective studies/ or retrospective studies/ or cohort.ti,ab. or longitudinal.ti,ab. or prospective.ti,ab. or retrospective.ti,ab. or quasi-experiment.ti,ab.

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36. Clinical Trial.pt. or random*.tw. or Clinical Trials/ or clinical trial.tw. or Follow-up Studies/ or prospectiv*.tw. or control*.tw.
37. or/34-36
38. 33 and 37

EMBASE

1. (TEVAR or EVAR).ti,ab.
2. (endovascular adj1 (stent* or repair* or treat* or surg* or (aneurysm? adj1 repair*))).ti,ab.
3. (open adj2 (surg* or repair*)).ti,ab.
4. endoprosthe*.mp.
5. Thoracic Aorta Aneruysm/su [Surgery]
6. Thoracoabdominal aorta aneurysm/su [Surgery]
7. Thoracic Aorta/su [Surgery]
8. exp Thoracic aorta surgery/
9. or/1-8
10. exp Thoracic Aorta Aneurysm/
11. exp Thoracoabdominal aorta aneurysm/
12. Thoracic Aorta/
13. (thorac* adj2 aneurysm*).ti,ab.
14. (descending adj1 aort* adj1 aneurysm*).ti,ab.
15. TAA*.ti,ab.
16. (arch adj3 aneurysm*).ti,ab.
17. or/10-16
18. (outcome? or mortalit*).mp.
19. complication*.mp.
20. ((technical or surg*) adj2 (advance* or improve*)).mp.

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21. (quality adj2 life).mp.
22. Mortality/
23. Postoperative complication/
24. peroperative complication/
25. Thoracic Aorta Aneurysm/co
26. “Quality of Life”/
27. or/18-26
28. (Randomised Controlled Trial or Pragmatic Clinical Trial).pt or exp Randomized Controlled Trials as Topic/ or “Randomized Controlled Trial (topic)”/ or Randomized Controlled Trial/ or Randomization/ or Random Allocation/ or Double-Blind Method/ or Double Blind Procedure/ or Double-Blind Studies/ or Single-Blind Method/ or Single Blind Procedure/ or Single-Blind Studies/ or Placebos/ or Placebo/ or (random* or sham or placebo*).ti,ab,hw,kf,kw. or ((singl* or doubl*) adj (blind* or dumm* or mask*)).ti,ab,hw,,kw. or ((tripl* or trebl*) adj (blind* or dumm* or mask*)).ti,ab,hw,,kw.
29. random:.tw. or placebo:.mp. or double-blind:.mp.
30. controlled clinical trial/ or clinical trial/ or exp controlled study/ or randomized controlled trial/ or exp quasi experimental study/
31. or/28-30
32. 9 and 17 and 27 and 31
33. limit 32 to (human and yr=”1994-Current”)
34. exp cohort analysis/ or exp longitudinal study/ or exp prospective study/ or exp follow up/ or cohort*.tw.
35. limit 34 to (human and yr=”1994-Current”)
36. 9 and 17 and 27 and 35
37. 36 not 33
38. 33 or 37

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Appendix 2: Data extraction table

Study	**Von Allmen et al 2014**	**Goodney et al 2011** (2 analyses*)	**Gore TAG**	**Hughes et al 2014**	**Piffaretti et al 2007**
METHODS					
Study Design	Comparative cohort	Comparative cohort	Comparative cohort	comparative cohort	Comparative cohort
Method of matching	None	1. None 2. propensity matched	ESG patients judged eligible for OSR	None	OSR patients judged eligible for ESG
Inclusion Criteria	Hospital diagnosis code 171.2 (TAA without mention of rupture) from ICD-10 score linked to	Patients diagnosed with a thoracic aneurysm according to ICD9 diagnosis code:	Fusiform descending thoracic aortic aneurysm at least twice the size of the normal adjacent aorta or saccular	Nationwide inpatient Sample database of patients >18 years old who had	Elective, Chronic fusiform or saccular aneurysm of the descending thoracic aorta, OSR patients

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	operations code according to OPCS-4, linked with all-cause mortality from ONS and ICD-10, only OPCS-4 codes: L18.2, L19.2, L20.2, L21.2	Intact thoracic aneurysms 441.2 Patients had undergone an open or TEVAR procedure according to ICD9 procedure codes: Open 38.35, 38.45 TEVAR 39.73, 39.79	aneurysm, Life expectancy >2 years Surgical candidate Male or infertile female >21 years old Descending aorta must be clampable distal to the left carotid artery, and distal anastomosis must be performed proximal to the celiac axis Inner aortic diameter of 23-37mm adjacent to aneurysm Lack of significant thrombus of calcification in landing zones	undergone an open or endovascular repair of non-ruptured thoracic aortic aneurysm. Patients selected using ICD-9 procedures codes 39.73 and 38.45.	would have been a candidates for ESG due to morphological criteria, Selected on surgeon's opinion of lower risk of ESG

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			Minimum of 2cm of normal thoracic aorta proximal and distal to aneurysm Aortic taper of no more than 4mm or ability to treat with more than one graft		
Exclusion Criteria	Dissection, Abdominal aneurysms, Non-identified area aneurysm, Ascending aneurysm, Suprarenal aneurysm, Non-	Patients with ICD 9 diagnosis codes indicating ascending thoracic aneurysms or cardiopulmonary bypass occurring with circulatory arrest. Thoracoabdominal aneurysm, thoracic aortic	Mycotic aneurysm, Hemodynamically unstable ruptured aneurysm , Major operation (other than planned subclavian to carotid transposition or bypass) within 30 days, MI or CVA within 6 weeks, Creatinine >2.0mg/dL, Connective tissue	NA	Traumatic, Mycotic thoracic aneurysms, Dissections, Aneurysms of the ascending aorta, Aneurysms of the aortic arch, Aortic ulcers

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	aortic procedures, age <50yrs	dissection, other aortic pathology. IC9 procedure codes indicating debranching or other procedures to extend endovascular landing zones (39.24, 39.25,	disorder, Acute or chronic aortic dissection, Planned occlusion of carotid or celiac arteries, Documented drug abuse within 6 months, Participation in another investigational device or drug study within 1 year		
Definition of the short-term	30 days post-procedure	latest occurring of discharge or 30 days post-intervention	latest occurring of discharge or 30 days post-intervention	post-procedure hospital stay	30-days post-intervention
PARTICIPANTS					
Country	UK	USA	USA	USA	Italy

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Setting	Multi-centre – Hospital episode statistics	Multi-centre – Medicare data	Multi-centre	Multi-centre	Single-centre
Time frame	2006 to 2011	1. 1998-2007 2. 2003-2007	ESG 1999- 2001, OSR pre 1999 (n=44) & 1999-2001 (n=50)	1998-2007	ESG 2000-2007, OSR 1996-2000
Recruitment	Total 618 ESG 354 OSR 264	1. Total 13,998 (ESG 2433, OSR 11,565) 2. Total 1100 (ESG 550, OSR 550)	Total 234 ESG 140 OSR 94	Total 8967 ESG 712 OSR 8255	Total 28 ESG 17 OSR 11
Age summary	median (IQR) ESG 73(66,78), OSR 71 (63-76)	mean (95% C.I.) 1. ESG 75.9 (75.6, 76.1), OSR 73.8 (73.7, 73.9) 2. ESG 71.1 (70.8, 71.44), OSR 70.7 (70.7, 71.1)	mean(SD) ESG 75.9 (10.4), OSR 68.2 (10.2)	median (IQR) ESG 72 (63- 78), OSR 63 (52, 72)	mean (SD) ESG 66 (10), OSR 61 (13)

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Proportion male	ESG 232 (65.5%), OSR 137 (51.9%)	% (95% C.I.): 1. ESG 58.7 (56.7, 60.7), OSR 55.4 (54.8, 56.3) 2. ESG 64.0 (59.9, 68.0), OSR 68.1% (64.2, 72.0)	ESG 90(57%), OSR 48 (51%)	5415 (61.3%) ESG, 437 (65.6%) OSR	ESG 14 (82%), OSR 8 (73%)
Black race	NR	% (95% C.I.): 1. ESG 7.5 (6.4, 8.5), OSR 3.8 (3.4, 4.5) (p=0.001) 2. ESG 2.7 (1.4, 4.1), OSR 2.7 (1.4, 4.0) (p=0.99)	NR	NR	NR
Charlson comorbidity score	NR	% (95% C.I.): 1. ESG 1.46 (1.38, 1.54), OSR 0.83 (0.80, 0.86) (p=0.00001)	NR	NR	NR

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		2. ESG 0.5 (0.41, 0.60), OSR 0.44 (0.35, 0.53) (p=0.34)			
Diabetes	NR	% (95% C.I.): 1. ESG 8.00 (6.7, 8.8), OSR 4.10 (3.7, 4.5) (p=0.001) 2. ESG 4.90 (3.0, 6.7), OSR 2.90 (1.5, 4.3) (p=0.09)	NR	684 (13.2%) ESG, 94 (8.3%) OSR (p<0.001)	NR
Myocardial infarction	NR	% (95% C.I.): 1. ESG 7.00 (6.00, 8.1), OSR 5.10 (4.6, 5.4) (p=0.001) 2. ESG 2.50 (1.2, 3.8), OSR 3.40 (1.9, 4.9) (p=0.38)	-NR	NR	NR
Congestive heart failure	NR	% (95% C.I.):	ESG 13 (9%), OSR 9 (10%) (p=1.00)	NR	NR

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		1. ESG 10.85% (9.6%-12.0%), OSR 10.60% (10.1%- 11.2%) (p=0.16) 2. ESG 3.00 (1.6, 4.5), OSR 4.70 (2.9, 6.5) (p=0.16)			
Cerebrovascular disease	NR	% (95% C.I.): 1. ESG 5.09% (4.2%-5.9%), OSR 3.20% (2.9%- 3.5%) (p=0.65) 2. ESG 2.00 (0.5, 2.6), OSR 2.00 (0.8, 3.1) (p=0.65)	NR	NR	4 (23%) ESG, 1 (9%) OSR (p=NS)
COPD	NR	% (95% C.I.): 1. ESG 22.04% (20.3%-23.6%), OSR 10.65% (10.0%-11.2%) (p=0.47)	ESG 56 (40%), OSR 36 (38%) (p=0.89)	247 (34.7%) ESG, 1483 (17.9%) (P<0.001)	10 (59%) ESG, 6 (54%) OSR (p=NS)

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		2. ESG 4.00 (2.3, 5.6), OSR 4.90 (3.0, 6.7) (p=0.47)			
Chronic renal failure	NR	% (95% C.I.): 1. ESG 4.9% (4.0%- 5.7%), OSR 0.88% (0.07%- 0.10%) (p=0.70) 2. ESG 0.01 (0, 1.4), OSR 0.50 (0, 1.1) (p=0.70)	NR	90 (12.6%) ESG, 126 (1.5%) OSR (p<0.001)	4 (23%) ESG, 1 (9%) OSR (p=NS)
History of malignancy	NR	% (95% C.I.): 1. ESG 5.71% (4.7%-6.6%), OSR 2.91% (2.6%- 3.2%) (p=0.82) 2. ESG 1.80 (0.6, 2.9), OSR 1.60 (0.5, 2.6) (p=0.82)	NR	NR	NR
History of smoking (current or past)	NR	NR	ESG 117 (84%), OSR 77 (82%) (p=0.86)	NR	NR

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BMI (kg/m2)	NR	NR	mean (SD): ESG 26.4 (4.7), OSR 26.9 (5) (p=0.44)	NR	NR
Coronary artery disease	NR	NR	ESG 69 (49%), OSR 34 (36%) (p=0.60)	NR	NR
Stroke	NR	NR	ESG 14 (10%), OSR 9 (9%) (p=1.00)	NR	NR
Peripheral arterial disease	NR	NR	ESG 22 (16%), OSR 10 (11%) (p=0.33)	NR	NR
Symptomatic aneurysm	NR	NR	ESG 30 (21%), OSR 36 (38%) (p=0.007)	NR	NR
Other concomitant aneurysm:	NR	NR	ESG 39 (28%), OSR 26 (28%) (p=1.00)	NR	NR
Renal dialysis	NR	NR	ESG 2 (1%), OSR 0 (0%) (p=0.52)	NR	NR

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Paraplegia	NR	NR	ESG 1 (1%), OSR 0 (0%) (p=1.00)	NR	NR
Cancer	NR	NR	ESG 27 (19%), OSR 12 (13%) (p=0.21)	NR	NR
Prior major vascular surgery	NR	NR	ESG 63 (45%), OSR 52 (55%) (p=0.14)	NR	NR
Cardiac comorbidities	NR	NR	NR	265 (37.2%) ESG, 3535 (42.8%) (p<0.001)	NR
Hypertension	NR	NR	NR	NR	16 (94%) ESG, 10 (91%) OSR (p=NS)
Peripheral obstructive disease	NR	NR	NR	NR	5 (29%) ESG, 2 (18%) OSR (p=NS)
Ischemic heart disease	NR	NR	NR	NR	6 (35%) ESG, 5 (45%) OSR (p=NS)

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Cardiac Surgery	NR	NR	NR	NR	(18%) ESG, 0 (0%) OSR (p=NS)
SHORT- TERM OUTCOMES					
All-cause mortality	ESG 21 (5.9%) OSR 4 (1.5%) Adjusted OR: 0.79	1. ESG 6.1% (95%CI 5.1%, 7.0%) OSR 7.1% (95%CI 6.7%, 7.6%) 2. ESG 4.2% (95% C.I. 2.5%-5.8%) OSR 4.5% (95% C.I. 2.8%-6.2%)	ESG 3 (2.1%) OSR 11 (11.7%) Adjusted OR: 0.54	ESG 25 OSR 379	ESG 1 (6%) OSR 2 (20%)
Re-intervention	ESG 21 (5.9%) OSR 4 (1.5%) Adjusted OR: 2.97 (1.09, 8.10)	NR	NR	NR	NR

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Paraplegia/ Paraperisis	NR	NR	ESG 4 (3%) OSR 13 (14%)	NR	NR
Cerebral Vascular Accident	NR	NR	ESG 5 (4%) OSR 5 (4%)	NR	ESG 2 (9%) OSR 1 (12%)
Neurological- unspecified type	NR	NR	NR	ESG 20 (2.8%) OSR 273 (3.3%) Adjusted OR: 0.38	NR
Respiratory failure	NR	NR	ESG 5 (4%) OSR 19 (20%)	NR	NR
Pneumonia	NR	NR	NR	NR	ESG 2 (12%) OSR 3 (27%)
Pulmonary- unspecified type	NR	NR	NR	ESG 17 (2.4%) OSR 462 (5.6%) Adjusted OR: 0.48	NR
Myocardial Infarction	NR	NR	ESG 0 (0%) OSR 1(1%)	NR	ESG 0 (0%) OSR 1 (9%)

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Cardiac-unspecified type	NR	NR	NR	ESG 28 (2.9%) OSR 1252 (15.2%) Adjusted OR: 0.24	NR
Peripheral vascular disease	NR	NR	ESG 20 (14%) OSR 4 (4%)	NR	NR
Renal failure	NR	NR	ESG 2 (1%) OSR 12 (13%)	NR	NR
Wound infection/dehiscence	NR	NR	ESG 5 (4%) OSR 10 (11%)	NR	NR
GI complication	NR	NR	ESG 3 (2%) OSR 6 (6%)	NR	NR
Post-implant syndrome	NR	NR	NR	NR	ESG 4(18%) OSR 0(0%)
LONG-TERM OUTCOMES					

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All-cause survival	1 year adjusted OR: 1.1 95%CI [0.70-1.73], p=0.667 5 year adjusted HR: 1.45 95%CI [1.08-1.94], p=0.013	NR	NR	NR	NR
aortic-related survival	5 year adjusted HR: 1.27 95%CI [0.81-1.98], p=0.298	NR	NR	NR	NR
survival from aortic-related re-intervention	5 year adjusted HR: 1.70 95%CI [1.11-2.60], p=0.014	NR	NR	NR	NR
* 1. Unadjusted cohort, 2. Propensity matched cohort					

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Appendix 3: Risk of Bias assessment

Risk of Bias Results using the ROBINS-I risk of bias tool

Domain – source of bias	Goodney et al 2011 *(2 analyses*)*	Gore TAG	Hughes et al 2014	Piffaretti et al 2007	Von Allmen et al 2014
Confounding	1. Severe 2. Moderate	Severe	Severe	Severe	Severe
Selection of participants into the study	1. Low 2. Low	Low	Low	Low	Low
Classification of interventions	1. Low 2. Low	Low	Low	Low	Low
Deviations from intended interventions	1. Low 2. Low	Low	Low	Low	Low
Missing data	1. Low 2. Low	Low	Low	Low	Low
Measurement of outcomes	1. Low 2. Low	Low	Low	Low	Low
Bias in selection of the reported result	1. Moderate 2. Moderate	Moderate	Moderate	Moderate	Moderate
Overall	1. Severe 2. Moderate	Severe	Severe	Severe	Severe
* 1. Unadjusted cohort, 2. Propensity matched cohort					

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Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	6
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	6
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	7
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	7
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	8
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	8

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Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	8
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	8
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	9
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency ($e.g., I^2$) for each meta-analysis.	9

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	9
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	9
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	9
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	Table 1, appendix 2
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	14, appendix 3
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	14-19
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	14-19
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	14-19
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	NA

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DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	19-21
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	21
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	22
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	22

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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